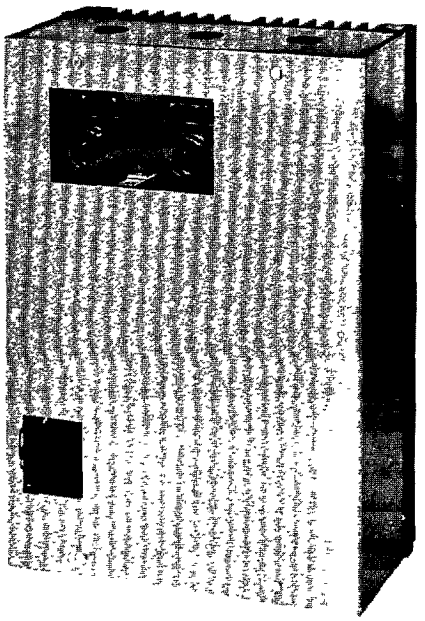
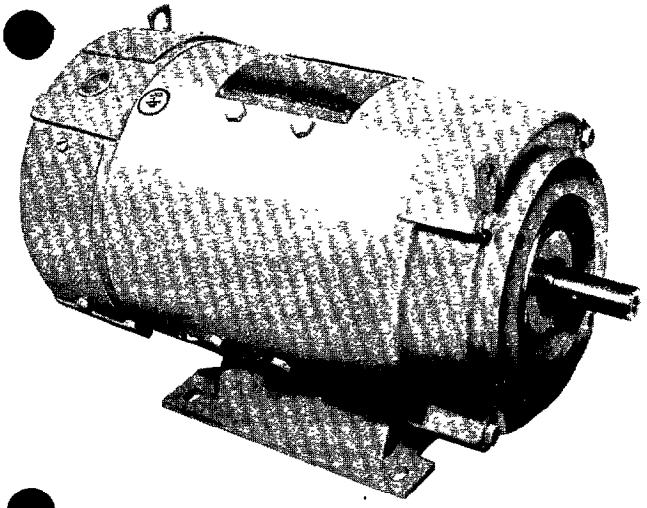
STATOTROL* II ADJUSTABLE SPEED DRIVE 6VFW SERIES

3 and 5 HP, SINGLE PHASE 230 VOLTS, 50/60 HERTZ





DEPENDABLE MOTOR CONTROL

Photo illustration shows controller with local operator's control installed

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company

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STATOTROL II DC DRIVE

INTRODUCTION

This manual contains general information on the General Electric Statotrol II DC drive. Additional information and diagrams may be supplied with the equipment when necessary.

DESCRIPTION

The Statotrol II drive is a high performance full wave DC drive. The regulating function is provided by a custom integrated circuit. The motor is a shunt wound field DC motor specially designed for use with a full wave phase controlled SCR power supply.

Standard features of the 3 and 5 HP Statotrol II drive include: NEMA-1 enclosure, speed control, protective current limit, undervoltage protection, and potentiometer adjustments for positive and negative IR compensation, maximum speed, and torque taper. Optional features include **dynamic braking, reversing, adjustable torque control, independently adjustable linear timed acceleration and deceleration, tach feedback, instrument follower, tach follower, speed indication, and a variety of special functions.**

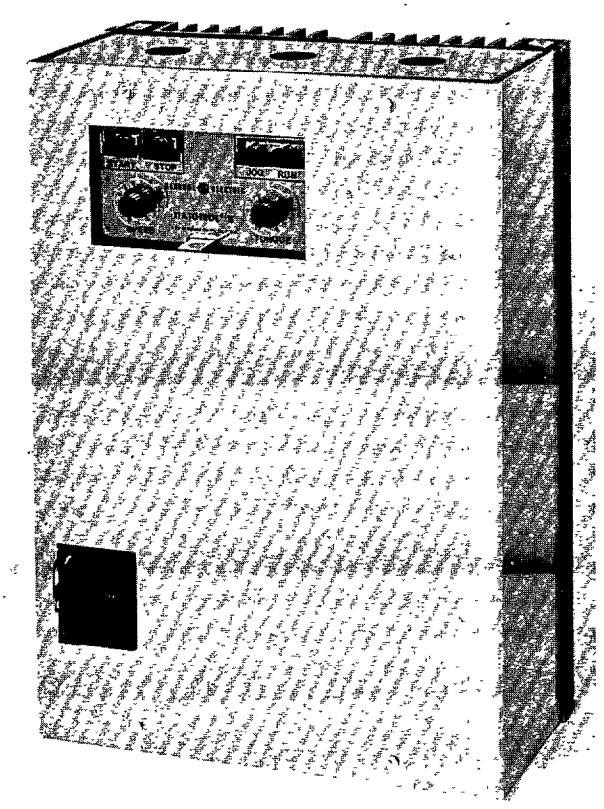


Figure 1. Statotrol II Motor Control 3 HP and 5 HP (with local operators control installed.)

RECEIVING AND STORAGE

As soon as the equipment is received, it should be unpacked and examined for any damage sustained in transit. If damage is evident, a damage claim should be filed immediately with the transportation company, and the local General Electric Co., Sales Office should be promptly notified.

If the equipment is not to be used as soon as it is unpacked, it should be stored in a clean, dry place and protected against accidental damage. Avoid storage in a location where construction work is in progress.

INSTALLATION

INSTALL CONTROL STATION AND OPTIONS

Either a local control station or a remote control station adapter unit must be attached to the main component board before operation is possible. Refer to Figure 2 for the mounting location and orientation of the control station. If a motor thermostat is to be used, and the controller has a local control station, the motor thermostat should be connected to the terminals on the control station component board before the control station is plugged into the controller.

When installing the control station use care to avoid bending any of the electrical connection pins. Be sure that each pin starts into its receptacle properly. Tighten the four mounting screws snugly to assure proper positioning of the control station.

Specific instructions for the installation and adjustment of the various options are on instruction sheets for the individual option. See "Modifications."

CONTROLLER INSTALLATION

The Statotrol II controller must be wall mounted in a location which will allow free flow of cooling air over the fins on the heat sink. Maximum ambient temperature around the controller must not exceed 40°C (104°F). Figure 3 gives mounting dimensions and outline dimen-

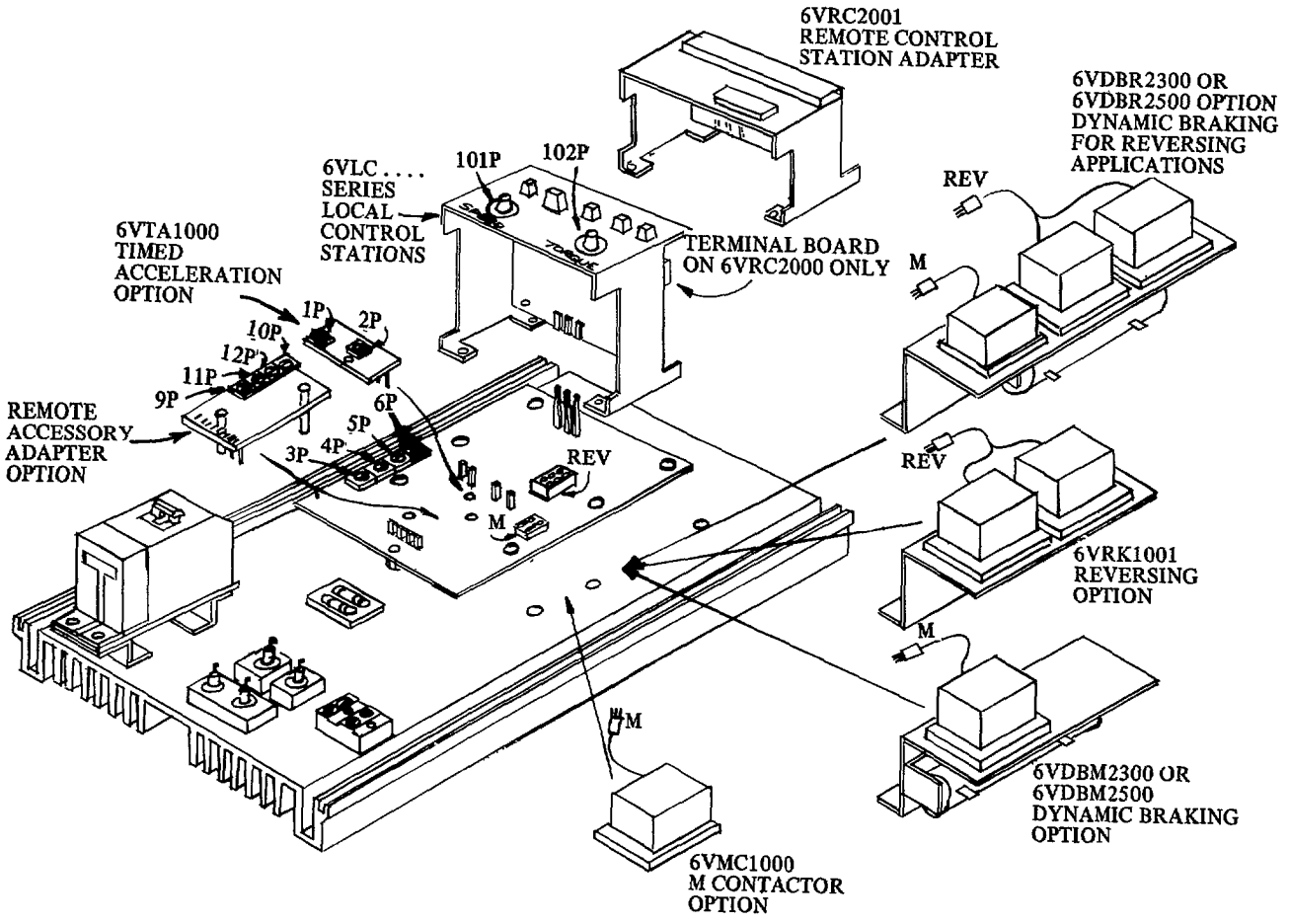
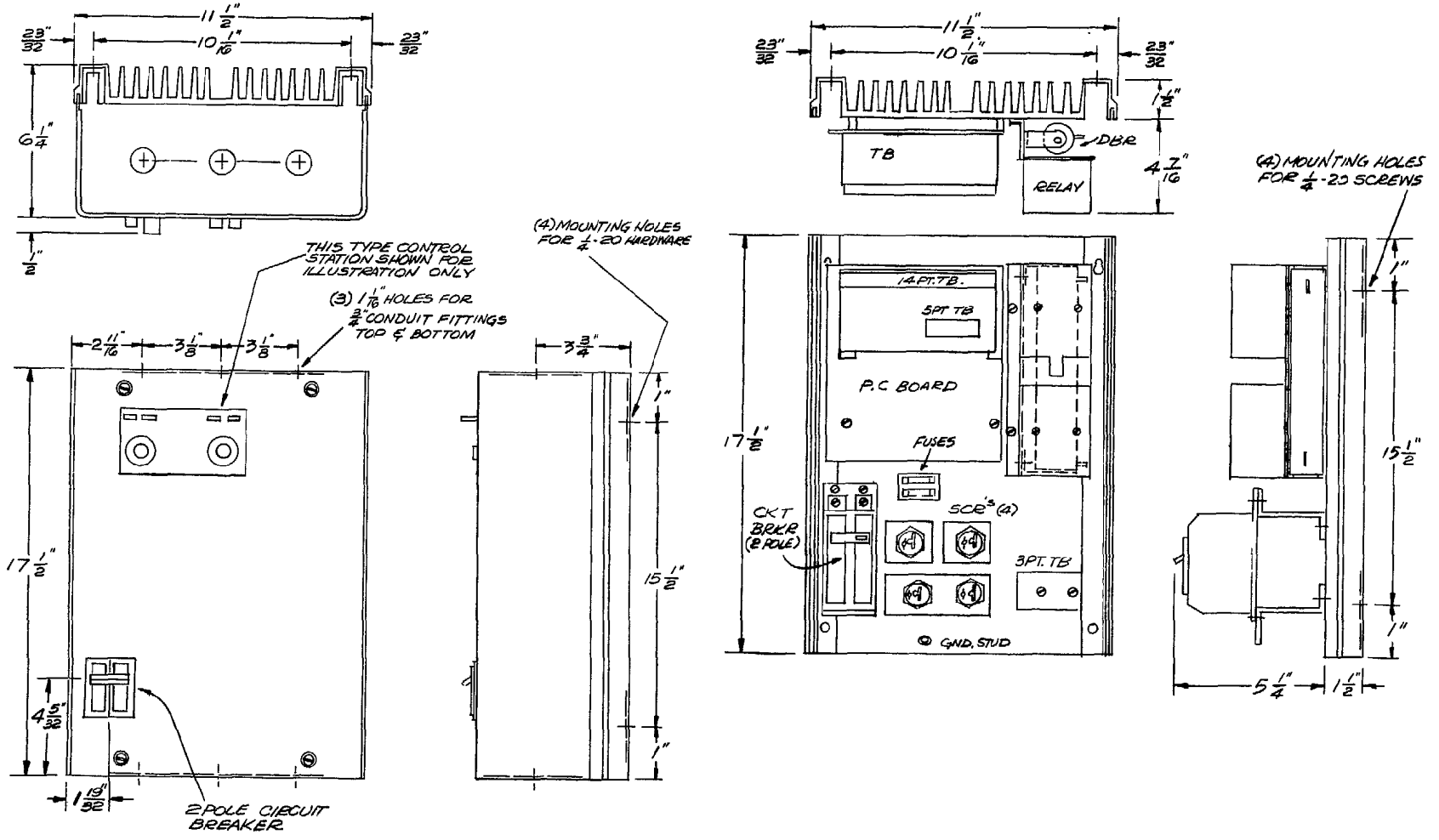


Figure 2. Exploded View of 3 HP and 5 HP Statotrol II Controller



6VFWB ENCLOSED VERSION

6VFWC CHASSIS MOUNT VERSION

Figure 3. Statotrol II Controller, 3 HP and 5 HP, Outline and Mounting Dimensions

sions. It is recommended that two or more inches of clearance be provided all around the controller to assure adequate air flow through the heat sink fins.

MOTOR INSTALLATION

The motor must be firmly mounted and properly aligned to prevent vibration. Excessive vibration causes rapid wear and objectionable audio noise. Heat dissipated by the motor will raise the ambient temperature around the motor if the motor is installed in an enclosed space. Since a lower ambient temperature will extend the motor life, it is always desirable to mount the motor in a well ventilated location. The motor should never be used where the ambient temperature exceeds 40°C 104°F) unless an oversized motor has been specially selected. Refer to the instruction book provided with the motor specific installation instructions.

When 5 HP TENV motors (258ATC or 259ATC frames) are coupled to C face gear reducers it may be necessary to support the motor weight with shims under the motor feet. Check the specifications on the reducer to be used.

WARNING

IF THE MOTOR IS ACCESSABLE WHILE IT IS RUNNING, A PROTECTIVE GUARD SHOULD BE INSTALLED AROUND ALL EXPOSED ROTATING PARTS.

ELECTRICAL CONNECTIONS

PROPER LINE VOLTAGE CONNECTION

This controller operates on 50/60 cycle 230 volts AC. A transformer is required where supply voltage deviates more than 10% from the controller rating. A calibration resistor 140R must be removed from the main component board if 50 Hz power is to be used. Refer to Figure 7.

GROUNDING

WARNING

PERSONNEL SAFETY CONSIDERATIONS AND THE NATIONAL ELECTRIC CODE REQUIRE THAT ELECTRICAL APPARATUS ENCLOSURES BE SOLIDLY CONNECTED TO BUILDING GROUND.

The necessary wire gauge of the ground wire is determined by the rating of the branch circuit breaker.

Connect a suitably grounded lead to the ground connection terminal provided in the Statotrol II enclosure near the circuit breaker.

WARNING

PROPER MOTOR GROUNDING IS ALSO ESSENTIAL FOR PERSONNEL SAFETY. DO NOT DEPEND UPON MOTOR MOUNTING BOLTS TO GROUND THE MOTOR FRAME. A 1/4 - 20 BOLT IN A TAPPED HOLE IN THE MOTOR END SHIELD IS PROVIDED FOR THE GROUND CONNECTION.

POWER WIRING

Connect 230V 50 or 60 Hz lines to the controller circuit breaker. The AC power must not vary from nominal voltage by more than 10%. Resistor 140R (see Figure 7) must be cut out of the circuit before 50 Hz power is connected. Table I gives minimum required circuit breaker and wire sizes.

Figure 4 is a wiring diagram for the power connections. When the motor is connected as shown in Figure 4, the motor will rotate CCW as viewed from the opposite shaft end. Turn off the circuit breaker and reverse leads A1 and A2 for clockwise rotation.

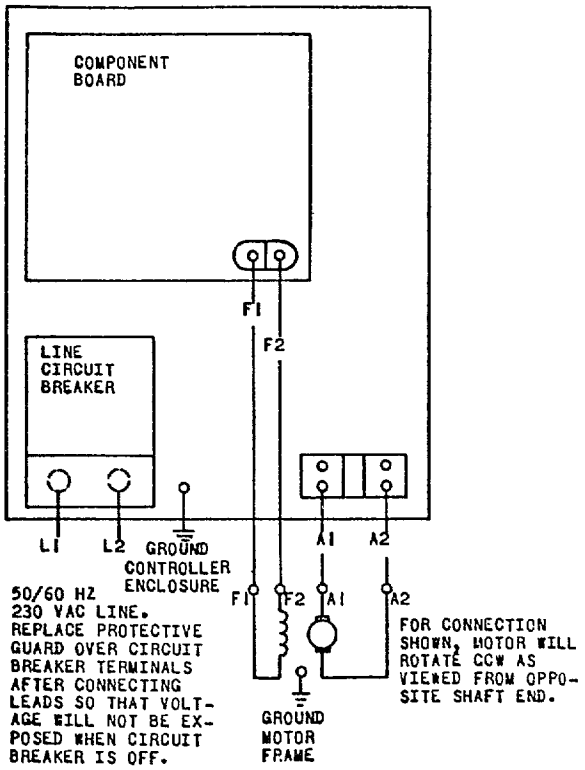


Figure 4. Power Wiring Connection Diagram

CAUTION

ANY CONNECTION ERROR IN THE POWER WIRING CAN DAMAGE THE CONTROL. RECHECK EACH CONNECTION BEFORE POWER IS APPLIED.

If there is any doubt about motor lead indentification, the problem can be resolved with a simple resistance check. The motor armature resistance will be about 0.3 ohms to 1 ohm, while field resistance will be about 200 ohms to 500 ohms. There must be no continuity between armature and field.

Refer to the instruction book packed with the motor for specific motor connection instructions. If the motor has leads marked C1 and C2, these leads should be individually taped, well insulated and left unconnected.

The branch circuit breaker must be large enough to eliminate nuisance tripping and small enough to protect the branch circuit and motor connection conductors. Refer to the latest revision of the National Electric Code and Table I on this page for minimum circuit breaker and wire sizes. Wherever local codes are more restrictive, the local codes take precedence over the National Electric Code. Special requirements exist for installation in hazardous locations and other special situations.

A line power isolation transformer must be used when the Statotrol drive is connected to an instrument signal source which is not isolated from ground. An individual line power isolation transformer **must** be applied when more than one Statotrol drive follows a commmon reference signal. When a line power transformer is used, any line switch used to disconnect the controller should be on the load side of the transformer.

REMOTE CONTROL STATION WIRING

When a remote control station is to be installed, it must be connected to 6VRC2000 or 6VRC2001 series adapter unit which plugs into the Statotrol II controller main component board. The connections for the forward, reverse, off, start, jog, run, and stop buttons are Class II control circuits as defined in Article 725 of the National Electric Code (1975).

Conductors connected to the speed adjustment potentiometer, the torque adjustment potentiometer, and the auto-manual switch meet the Class I control and signal circuit requirements of Article 725 of the National Electric Code.

TABLE I

STANDARD DRIVES HORSEPOWER RATING	MINIMUM WIRE AND CIRCUIT BREAKER SIZE
3 Horsepower	30 AMP Circuit Breaker 75°C #10 AWG copper wire for line and armature 75°C #14 AWG copper wire for motor field
5 Horsepower	45 AMP Circuit Breaker 75°C #8 AWG copper wire for line and armature 75°C #14 AWG copper wire for motor field

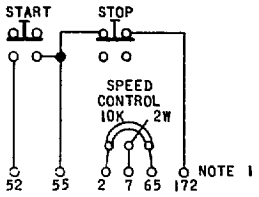
Figure 5 shows various remote control station connection schemes which can be used. To avoid electrical noise pickup, it is necessary to keep the torque potentiometer, speed potentiometer and auto-manual switch wires separate from all wires. Do not run these wires through conduits with power conductors or relay coil wiring. All AC brakes and relay coils should be suppressed.

MOTOR THERMOSTAT CONNECTIONS

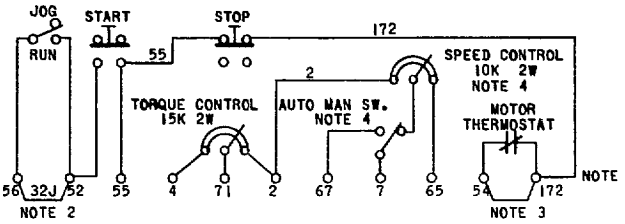
When a motor thermostat is used, it should be connected in series with the "off" switch. Refer to Figure 5 for connection diagrams of remote control stations with motor thermostats. Local control stations (those which are inside the controller enclosure) have a jumper across two terminals on the foil side of the control station component board. Remove the jumper and connect the thermostat leads across these two terminals. Use insulated lugs on the thermostat leads.

REMOTE ACCESSORY WIRING

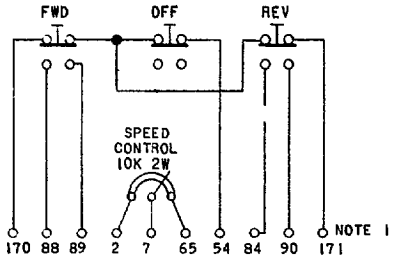
A seven point terminal strip, 2TB, is provided for connection of leads from remote accessories. This terminal strip is not used unless one of the optional remote accessory adapter units has been installed in the Statotrol controller. (See Figure 2). Remote accessory functions include tachometer feedback, tachometer follower, instrument follower and speed indication. Any combination of these functions can be used simultaneously with the exception of the tachometer follower and the instrument follower which are mutually exclusive.



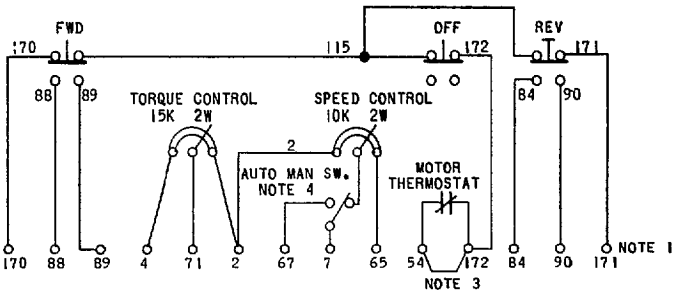
SIMPLEST NON-REVERSING REMOTE CONTROL STATION



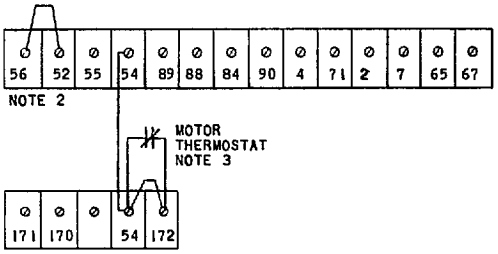
FULL OPTION NON-REVERSING REMOTE CONTROL STATION. ANY SINGLE FEATURE OR ANY COMBINATION OF TORQUE CONTROL, JOG-RUN, AUTO-MANUAL, AND MOTOR THERMOSTAT MAY BE CONNECTED.



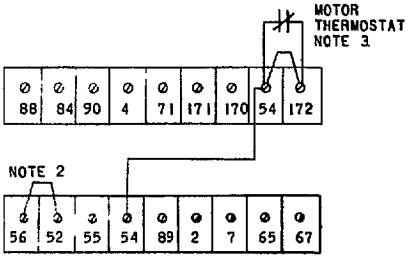
SIMPLEST REVERSING REMOTE CONTROL STATION



FULL OPTION REVERSING CONTROL STATION. ANY SINGLE FEATURE, OR ANY COMBINATION OF TORQUE CONTROL, AUTO-MANUAL, AND MOTOR THERMOSTAT MAY BE CONNECTED. FOR JOG-RUN SEE NOTE 8.



TERMINAL ARRANGEMENT ON 6VRC2001 REMOTE CONTROL STATION ADAPTER



TERMINAL ARRANGEMENT ON 6VRC2000 REMOTE CONTROL STATION ADAPTER

NOTES

1 TERMINAL ARRANGEMENTS ON SCHEMATIC DIAGRAMS DO NOT REPRESENT ACTUAL TERMINAL LOCATION, SEE SKETCHES AT LEFT FOR ACTUAL TERMINAL ARRANGEMENT ON REMOTE CONTROL STATION ADAPTERS

2 FACTORY INSTALLED METAL JUMPER FROM TERMINAL 56 TO TERMINAL 52 MUST BE REMOVED BY CUSTOMER WHEN JOG-RUN SWITCH IS CONNECTED

3 FACTORY INSTALLED METAL JUMPER FROM TERMINAL 54 TO TERMINAL 172 MUST BE REMOVED WHEN MOTOR THERMOSTAT IS CONNECTED

4 GOLD PLATED CONTACTS ARE REQUIRED ON SWITCHES OR RELAYS USED FOR SWITCHING SPEED CONTROL POTENTIOMETER OR TORQUE CONTROL POTENTIOMETER.

5. FORWARD, OFF, REVERSE, STOP AND START SWITCHES ARE MOMENTARY, AUTO-MANUAL AND JOG-RUN SWITCHES ARE MAINTAINED

6 CONTROL STATION WIRING MUST ENTER CONTROLLER ENCLOSURE THROUGH CONDUIT HOLES AT TOP OF CASTING.

7 WHEN FORWARD, REVERSE, OR START SWITCHES WITH MAINTAINED CONTACTS ARE TO BE USED, A 6VRC2002 OR 6VRC2003 REMOTE CONTROL STATION ADAPTER MUST BE USED, WIRE PER "MODIFICATION INSTRUCTION 6VRC2002 AND 6VRC2003 REMOTE CONTROL STATION" IN THIS MANUAL

8. A MAINTAINED CONTACT ADAPTER, 6VRC2002 OR A 6VRC2003, IS REQUIRED WHEN JOG-RUN IS TO BE USED WITH REVERSING

Figure 5. Remote Control Station Connections

ception of the tachometer follower and the instrument follower which are mutually exclusive.

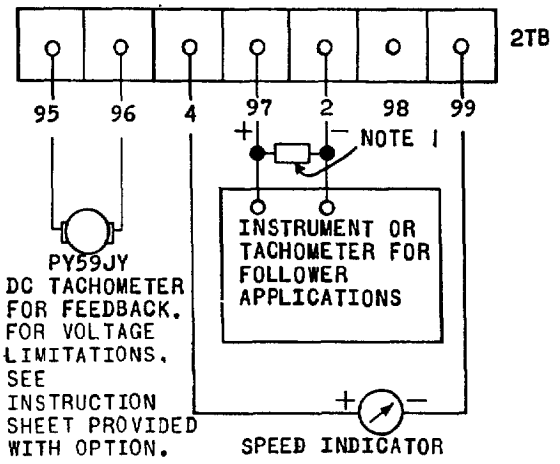
CAUTION

ALL CONNECTIONS TO REMOTE ACCESSORY OPTION MUST BE PHYSICALLY SEPARATED FROM ALL OTHER WIRES. IF THIS IS NOT FEASIBLE, SHEILDED WIRE MUST BE USED. CONNECT SHIELD LEAD TO 2TB(2). DO NOT GROUND SHIELD. SHIELD MUST BE INSULATED TO PREVENT ACCIDENTAL CONTACT TO BUILDING GROUND.

Figure 6 shows the wiring connections for remote accessories. The wiring shown should be considered as Class I signal wiring (National Electric Code, 1975, Article 725), except the instrument signal wiring which may be considered Class II if the instrument output is limited as specified in the referenced code article.

The tachometer feedback function is installed by coupling a PY59JY DC tachometer to the motor and connecting the tachometer leads to terminal strip 2TB as shown in Figure 6. The polarity of the signal and the direction of rotation do not affect this connection. The recommended PY59JY tachometer has only two terminals. (The identification numbers on the tachometer terminals do not correspond to the controller terminal numbers.) One of the optional remote accessory adapter units, 6VFB series, must be installed in the Statotrol II controller (refer to figure 2) to complete the circuit. Refer to the instruction sheet for the individual option for complete instructions on installation, adjustment, and voltage limitations.

The tachometer follower signal is obtained from a PY59-JY DC tachometer by connecting the tachometer leads to points 97 and 2 on terminal strip 2TB as shown in Figure 6. The positive side of the tachometer must be connected to terminal 97. Terminal number 2 on the PY59JY tachometer (tachometer terminal numbers do not correspond to the controller terminal numbers) is positive when the tachometer rotates counterclockwise as viewed from the opposite shaft end. The polarity reverses when the direction of rotation reverses. The output voltage of the tachometer must be limited (refer to the instruction sheet for the individual option for maximum and nominal voltages). One of the optional remote accessory adapter units 6VFB1000 or 1001 or 6VFT1000 or 1001, must be installed in the Statotrol controller (refer to Figure 2) to complete the circuit. When a local control station is used, it must be one of the following four: 6VLC1003, 6VLC1005, 6VLC1009, or 6VLC1011. Refer to Figure 5 for connections for remote control stations. Refer to the instruction



NOTE 1.

Protective shunt resistor is required to prevent overvoltage when a current signal instrument with maximum output of more than 5 milliamps is used. Use a 560 ohm resistor with a 20 ma max. signal, and use a 180 ohm resistor with a 50 ma max. signal. The proper value of the protective shunt resistor can be found by the following formula when instruments with other maximum outputs are used.

$$R = \left(\frac{8500}{I - 5}\right) \text{ ohms } \pm 10\%$$

R is the resistance of the protective shunt resistor and I is the maximum output of the signal generator in milliamps.

NOTE 2:

See instructions for individual option for connection diagram when instrument with voltage signal output is used in follower applications.

Figure 6. Remote Accessory Connection Diagram

page for the remote accessory adapter unit for complete instructions on installation and adjustment.

The instrument follower function is installed by connecting the instrument signal to points 97 and 2 on terminal strip 2TB as shown in Figure 6. The positive terminal of the signal generator must be connected to terminal 97. One of the optional remote accessory adapter units, 6VFB-1100, 1101, or 6VMA1000, must be installed in the Statotrol II controller (refer to Figure 2) to complete the circuit. When a local control station is used, it must be one of the following four: 6VLC1003, 6VLC1005, 6VLC1009, or 6VLC1011. Refer to Figure 5 for connections for remote control stations.

CAUTION

WHEN THE INSTRUMENT SIGNAL IS NOT COMPLETELY ISOLATED FROM GROUND, IT IS NECESSARY TO USE A LINE POWER ISOLATION TRANSFORMER TO SUPPLY THE STATOTROL DRIVE.

CAUTION

WHEN THE MAXIMUM CURRENT OUTPUT OF THE INSTRUMENT SIGNAL GENERATOR MAY EXCEED 5 MA, IT IS NECESSARY TO CONNECT A PROTECTIVE SHUNT RESISTOR FROM TERMINAL 97 TO TERMINAL 2 ON 2TB. FAILURE TO DO SO MAY RESULT IN DAMAGE TO THE STATOTROL II CONTROL CIRCUITRY. SEE FIGURE 6, NOTE 1, FOR PROPER RESISTANCE VALUE.

PREPOWER CHECKS AND ADJUSTMENTS

WARNING

THIS SECTION CONTAINS IMPORTANT WARNINGS AND CAUTIONS WHICH MUST BE OBSERVED DURING INSTALLATION OF THE CONTROL AND MOTOR. FAILURE TO OBSERVE THESE CAUTIONS MAY CAUSE SAFETY HAZARDS OR EQUIPMENT DAMAGE. READ THIS SECTION CAREFULLY AND MAKE ALL NECESSARY ADJUSTMENTS BEFORE POWER IS APPLIED TO THE CONTROL.

CURRENT LIMIT SETTING

The current limit circuit is factory adjusted for low impedance motors. If a medium or high impedance motor is to be used, the current limit must be adjusted at the time of installation. To set the current limit, first refer to Table II and locate the motor which is to be used.

After finding the relative impedance of the motor to be used, refer to Figure 7 for instructions on how to position the current limit setting jumper.

50 CYCLE POWER ADJUSTMENT

The controller is factory adjusted for use with 60 Hz AC power. If 50 Hz power is to be used, resistor 140R must be cut out of the circuit before 50 Hz power is applied. See Figure 7 for the location of 140R.

MOTOR CHECKS

If the motor is accessable while it is running, a protective guard should be installed around all exposed rotating parts.

The "Inspection Before Starting" section of the motor instruction book must be carefully followed.

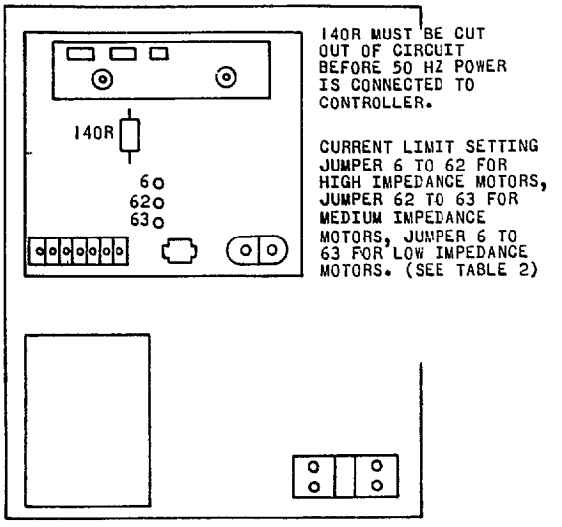


Figure 7. Current Limit Jumper Setting and 50 Hz Adjustment

TABLE II
CURRENT LIMIT SETTINGS FOR
3 HP and 5 HP MOTORS WITH
"6VFM . . . . ." CATALOG NUMBER
ON NAMEPLATE

All motors which do not have "6VFM . . . . ." number on nameplate and all 1150 RPM and all 3500 RPM motors.	See special instructions provided with control or consult factory. **Call (814) 455-0521**
3 HP and 5 HP 2500 RPM dripproof	high impedance motors (jumper 6 to 62)
3 HP and 5 HP 1750 RPM dripproof and 3 HP 2500 RPM TENV	medium impedance motors (jumper 62 to 63)
3 HP and 5 HP 1750 RPM TENV and 5 HP 2500 RPM TENV	low impedance motors (jumpers 6 to 63)

TABLE III

PREPOWER WIRING CHECKS

TERMINALS	APPROPRIATE RESISTANCE
Load side of open circuit breaker	**300 to 1000 ohms**
F1 to F2	200 to 500 ohms
A1 to A2	0.3 to 1.0 ohms
F1 to A1 or F2 to A2	500,000 ohms or more
Control Circuit to Ground	500,000 ohms or more (see exceptions in preceding paragraphs).

WIRING CHECK

Any connection error in the power wiring can damage the control as soon as power is applied. Any short to building ground in the signal wiring can cause damage as soon as power is applied, unless the Statotrol II drive is connected to a line power isolation transformer. If the controller is connected to any instruments or equipment which may be grounded, a line isolation transformer must be used. In no case should the circuit be grounded at more than one point. If the drive is to be operated as an instrument follower, verify that the correct protective resistor is properly installed. (Refer to Figure 6.) It is advisable to recheck all wiring before power is applied. Make certain that all screw terminals are tight. If an ohmmeter is available, the simple resistance checks shown in Table III should be performed to verify that the power wiring is correct.

The relay coil circuit is normally grounded at circuit point 51, and is isolated from the other controller circuits.

GROUNDING

Safety considerations and the National Electric Code require the motor frame and the controller enclosure to be connected solidly to building ground. Do not rely on mounting bolts for grounding. Refer to the "Electrical Connections" section of this instruction book for ground lead connections.

INITIAL OPERATION AND ADJUSTMENT

When the control has been mounted and connected, and the prepower checks and adjustments are complete, apply power to the control, turn the speed control knob to zero, and press the "Start" button on the control station. Slowly turn the speed control knob until the motor starts to run. Check the direction of motor rotation to be sure it is correct. If the motor turns the wrong way, disconnect power from the control and reverse the motor armature leads.

The "Inspection After Starting" section of the motor instruction book should be carefully followed at this time.

The control is now ready for normal operation. Some Applications may require special settings for maximum speed, minimum speed, IR compensation, and torque taper. Potentiometers for adjusting these functions are in a line along one edge of the main component board inside the controller. Remove the controller cover and use a small screwdriver to adjust the potentiometers as described in the following instructions.

WARNING

HIGH VOLTAGE ELECTRIC SHOCK CAN CAUSE SERIOUS OR FATAL INJURY. BE CAREFUL NOT TO COME IN CONTACT WITH LIVE PARTS. LINE VOLTAGE IS EXPOSED IF THE LINE CIRCUIT BREAKER IS ON AND THE COVER IS REMOVED. ALWAYS TURN OFF THE CIRCUIT BREAKER BEFORE PERFORMING ANY WORK OTHER THAN POTENTIOMETER ADJUSTMENTS INSIDE THE CONTROLLER. INTERNAL ADJUSTMENTS SHOULD BE MADE BY QUALIFIED ELECTRICIANS AND NOT BY MACHINE OPERATORS.

MINIMUM SPEED ADJUSTMENT (4P)

The minimum speed adjustment is factory set so that the motor will start to run when the speed control knob is turned just slightly off zero. To change this setting, first turn the speed control knob to zero and then rotate the minimum speed potentiometer until the motor runs at the desired minimum speed.

The minimum speed adjustment is also used to match proper motor speed to specific signal levels when the reference signal comes from a signal generating instrument. This function is explained on the instruction page for the remote accessory adapter option boards.

Adjusting the minimum speed changes the maximum speed also, so the minimum speed should be set before the maximum speed potentiometer is adjusted.

MAXIMUM SPEED ADJUSTMENT (3P)

CAUTION

DO NOT CHANGE THE FACTORY SETTING OF

THE MAXIMUM SPEED POTENTIOMETER IF TACHOMETER FEEDBACK IS TO BE USED. IF THE MAXIMUM SPEED POTENTIOMETER IS INADVERTENTLY DISTURBED, IT SHOULD BE RETURNED TO APPROXIMATELY 80% OF ITS POSSIBLE CW ROTATION.

THE MAXIMUM SPEED ADJUSTMENT DOES NOT LIMIT MOTOR SPEED WHEN THE CONTROL IS IN THE "AUTOMATIC" MODE OF OPERATION (TACHOMETER FOLLOWER AND INSTRUMENT FOLLOWER APPLICATIONS).

The maximum speed adjustment is factory set to a speed slightly higher than rated motor speed. To change the maximum speed, first turn the speed control knob to its highest speed setting. Then adjust the maximum speed potentiometer until the motor runs at the highest speed needed for the particular application. The motor should never be operated above its rated speed.

IR COMPENSATION ADJUSTMENT (6P)

Conventional IR compensation boosts the motor speed at high load. The Statotrol II IR compensation circuit provides this feature and can also be adjusted to provide "droop" for applications which require motor speed to drop off linearly as load is added. The IR compensation adjustment factory setting is adequate for most applications. When necessary, the IR compensation can be adjusted by the following procedure. Start the drive and set the speed control knob to the speed at which regulation is most critical in your particular application. Adjust the motor load to minimum and then measure the motor speed precisely with a hand tachometer or a strobe light. NOTE: The speed of integral hp motors can be conveniently read by removing the dust cap on the commutator-end motor bearing and using a hand tachometer. On fractional hp motors, speed can be read with a strobe light at any convenient point where the rotating shaft can be seen.

Now adjust the motor load to maximum (not exceeding 100 percent rated torque) and again read motor speed. If the "maximum-load" speed is less than the "mimimum-load" speed, turn the IR compensation potentiometer clockwise (CW) until they are equal.

There is no need to wait for the motor to warm up since the Statotrol II IR compensation is relatively insensitive to motor temperature. Some loads may "hunt" if the IR compensation is set too high. Turning the IR compensation potentiometer CCW should eliminate this hunting.

TORQUE TAPER ADJUSTMENT (5P)

CAUTION

THE TORQUE TAPER ADJUSTMENT IS FACTORY SET FULL CCW. THE TORQUE TAPER ADJUSTMENT MUST BE KEPT FULL CCW WHEN A CONTROL STATION WITH A TORQUE CONTROL KNOB IS USED.

In applications which require a fixed program of torque control, the torque taper adjustment may be turned clockwise to increase the output torque which the motor will deliver at low speeds.

OPTION ADJUSTMENT

Detailed instructions are on the instruction page for the individual option. See "Modifications."

OPERATOR CONTROLS, NORMAL OPERATION

The operator controls have been made as simple and foolproof as possible. However, in certain applications, it may be necessary to caution the machine operator against operation sequences which may damage the machinery or process driven by the Statotrol motor. The following instructions apply only when the customer's load requires no special operating sequences.

STARTING

Press the "start", "forward", or "reverse" button. The speed control knob may be at any desired setting. The motor will accelerate smoothly to the speed set by the speed control knob.

Motor current is automatically limited to a safe value. No warmup is required, and motor response is immediate.

CHANGING SPEED

The speed control knob may be turned to any desired setting at any time, and the motor will respond smoothly. Turning the speed control knob clockwise will increase the motor speed.

STOPPING

Press the "stop" button. Motor will stop smoothly regardless of setting of knobs and other switches.

TORQUE CONTROL

The torque control knob adjusts the maximum output torque the motor will deliver. To operate in the torque control mode, first set the speed control knob to its highest desired speed and then turn the torque control knob to any desired setting.

Torque knob settings above 5 on the dial are for intermittent duty only, since settings above 5 may cause the motor to deliver more than rated torque (except with limited torque loads) and the motor will consequently overheat. The numbers on the torque control knob are for indication only, and should not be relied upon for motor overload protection. When the motor load is such that it will not overload the motor except during acceleration, the torque control knob may be left at any setting if the motor is not stopped and restarted continuously.

REVERSING

The rotation of the motor may be reversed at any time by pressing the "forward" or "reverse" button. The motor will smoothly reverse its direction of rotation. Operators should be cautioned to avoid rapidly cycling the motor from forward to reverse enough times to overheat the motor. (Proper setting of the torque control knob and the torque taper adjustment can prevent motor overheating by limiting armature current to the nameplate rating of the motor.)

JOG-RUN

To jog, push the "jog" button, and then the motor will run only while the operator holds the "start", "reverse", or "forward" button. When jogging is complete, push the "run" button and the drive will return to normal operation.

AUTO-MANUAL

When the "manual"button is set, the motor will respond to the speed control knob. When the "auto" button is set, the motor will respond to a speed reference signal from a remote signal source.

MAINTENANCE

CONTROLLER

The controller enclosure should be periodically inspected to prevent an accumulation of materials which might block the flow of cooling air through the heat sink fins. The control must be protected from dripping or sprayed moisture and oil.

MOTOR

Complete instructions for motor maintenance are included in the motor instruction book.

GEAR BOX

Periodic oil changes and bearing lubrication are necessary to prolong the life of most gear boxes. Refer to the instructions provided with the unit for recommended maintenance schedule and lubricants.

TACHOMETER

After 5 years of normal service or 2 years of severe duty the bearings should be inspected. If they appear to be loose or worn, they should be replaced. Brushes should be inspected after every 1000 hours of operation and replaced when they wear down to 3/8" in length.

TROUBLESHOOTING AND REPAIR

WARNING

HIGH VOLTAGE ELECTRIC SHOCK CAN CAUSE SERIOUS OR FATAL INJURY. BE CAREFUL NOT TO COME IN CONTACT WITH LIVE PARTS. LINE VOLTAGE IS EXPOSED IF THE LINE CIRCUIT BREAKER IS ON WHILE THE CONTROLLER COVER IS REMOVED. USE EXTREME CARE TO AVOID TOUCHING EXPOSED CONDUCTORS. ALWAYS TURN OFF THE LINE CIRCUIT BREAKER BEFORE DOING ANYTHING OTHER THAN ADJUSTING POTENTIOMETERS. THE CONTROL STATION STOP BUTTON DOES NOT REMOVE POWER FROM THE CONTROL OR THE MOTOR FIELD.

If a newly installed drive will not function properly, it is most likely that a terminal is loose, or a problem exists with a connection, line voltage, or an adjustment. Line voltage must be between 253 volts AC and 207 volts AC, and the adjustments must be set as described in the "Initial Operation and Adjustments" section of this book.

If the drive operates normally for a while and then malfunctions, the problem may be line voltage, motor overload, motor failure, a loose terminal, an open fuse, or a component failure. In the following discussions, each step of troubleshooting is based on the assumption that all preceding steps have been completed and nothing abnormal has been found.

SYMPTOMS AND THEIR PROBABLE CAUSES

Motor Will Not Run

First check the branch circuit breaker and the controller

circuit breaker to be certain they are closed. Then if the control station has one, turn the torque control knob clockwise. Then turn off the line circuit breaker and check fuses FU5 and FU6 in the controller.

WARNING

HIGH VOLTAGE ELECTRIC SHOCK CAN CAUSE SERIOUS OR FATAL INJURY. BE CAREFUL NOT TO COME IN CONTACT WITH LIVE PARTS. ALWAYS TURN THE CIRCUIT BREAKER OFF BEFORE REPLACING FUSES.

Replace any open fuse with one of the same type and amperage rating as the original fuse. Next, connect an AC voltmeter across the controller circuit breaker terminals and verify that the line voltage is between 207V and 253V AC. Next connect a DC voltmeter between field terminals F1 and F2 on the two point terminal strip on the component board and verify that the voltage is between 180 volts DC and 220 volts DC. If the field voltage is less than 150 volts DC, one of the diodes in the field power supply may have failed. These diodes are 16D, 17D, 18D and 19D and are located on the main component board just above the field connection terminals. If the voltage is zero, two more diodes may have failed, or a fuse may be open. If the field voltage is correct, press the start button, verify that the control relay on the component board and the M contactor (if used) pick up, turn the speed control knob to full speed, and read the DC voltage across terminals A1 and A2. If this voltage is 10 to 20 volts or more, the motor is stalled due to an overload, or there is an open in the motor leads, windings, or brushes. If the voltage from A1 to A2 is **zero, turn the minimum speed adjustment (4P) clockwise.** If the motor now starts, the problem is in the control station and its plug-in contacts should be checked and then the control station should be replaced. If the motor does not start, locate the two small connectors which are used to plug the SCR gate leads onto the main component board. Partially extract and then reseat each connector to restore electrical contact which may have been lost due to corrosion. If the motor still will not start, the main component board should be replaced.

Motor Runs At High Speed and Cannot be Controlled

If the drive system is used as a tachometer follower or an instrument follower, try to operate the control in the manual mode. If it works OK in manual, the problem is probably in the tachometer or instrument, or the "Remote Accessory Adapter" option (see Figure 2).

If the drive has tachometer feedback, remove the remote accessory adapter board (see Figure 2), jumper between the two option connection receptacles 22 and 20 as indicated by the yellow line on the main component board, and retest.

If the control has the "timed acceleration and deceleration" option (see Figure 2), disconnect the option. Jumper between the two receptacles 7 and 10 as shown on the main component board. (An explanatory note on the main component board is exposed when the options removed). Reconnect the power and retest.

If the motor speed now responds to the "minimum speed" or "maximum speed" adjustments, there is a short circuit in the control station. If the motor still runs at top speed only, replace the main component board.

Motor Runs Very Fast for Speed Setting, but Very Little Torque is Produced:

Just above 2TB on the main component board there is a row of receptacles which accept the pins of the Remote Accessory Adapter Option. If the option is not used, a jumper must be used to connect receptacles 20 and 22. The proper jumper location is marked by a yellow line on the component board. Verify that this jumper is in place if the option board is not being used. Next, check the AC line voltage at the controller circuit breaker terminals. Verify that this voltage is within $\pm$ 10% of the nominal 230V AC. Check the DC voltage from F1 to F2 on 1TB. If this voltage is less than 150V DC, one or more of the field power supply diodes has probably failed. These diodes are labeled 16D, 17D, 18D, and 19D, and are located on the main component board just above the field connection terminals.

Motor Operates Normally at No Load, but Will Not Deliver Adequate Torque to Drive Load

If the drive has operated properly in the past and suddenly develops this symptom, locate the two small connectors which are used to plug the SCR gate leads onto the main component board. Partially extract and then reseat each connector to restore electrical contact which may have been lost due to corrosion. Then, if an oscilloscope is available, connect it across terminals A1 and A2, and if full wave rectified power (a voltage pulse every 8-1/3 milliseconds) is being applied to the armature, reset the current limit as explained in the next paragraph. If no oscilloscope is available, or if the oscilloscope shows that half wave rectified power (a voltage pulse every 16-2/3 milliseconds) is being applied to the armature, replace the main component board and if the problem still persists, replace the SCR's. If using an oscilloscope make sure that it's chassis is isolated from ground.

If a newly installed drive will not deliver adequate torque to drive the load, first verify that the torque control knob (if used) is turned clockwise, and also verify that the motor is not running faster than its rated speed. Then turn off the line circuit breaker and connect ammeters in series with the motor armature and run the motor under load. If

the DC armature current exceeds the motor nameplate rating or if the AC armature current is more than 1.5 times the nameplate rating, the motor is overloaded. If the measured DC current is less than the motor nameplate rating, and if the AC current is less than 1.5 times nameplate rating turn off the line circuit breaker and reset the circuit limit adjustment to the setting appropriate to the next higher impedance group of motors in Table II and retest.

Circuit Breaker Trips

If the circuit breaker trips within a few seconds after power is applied, there is probably a short circuit or a wiring problem. Refer to the "Prepower Checks and Adjustments" section of this book for the appropriate wiring and impedance checks. If the input impedance is low, set the ohmmeter to the X1000 scale and measure the resistance between the cathode and anode (i.e., power terminal and heatsink) of each SCR. The meter should show open circuit for either polarity of the probes. If the reading on any SCR is not open circuit, that SCR should be replaced.

If the circuit breaker trips when the motor starts, turn off the line circuit breaker, refer to Table II and Figure 7, and reset the current limit adjustment to the setting appropriate to the next lower impedance group of motors. (If the control is being used with 50 Hz power, verify that resistor 140R has been cut out of the circuit as shown on Figure 7 before resetting the current limit.)

If the circuit breaker trips after a few minutes or a few hours of steady running, the motor is probably overloaded. Measure the DC armature current in the motor, and if it exceeds the motor nameplate rating, the motor is overloaded.

Fuse Blows

The instrument fuses 5FU and 6FU provide overcurrent protection for the component board, including the motor field DC power supply. If a fuse blows it should be replaced with a new fuse of the same type and rating. If the fuse blows again, turn off the line circuit breaker and measure the line to line resistance on the load side of the fuse block. This resistance should be 500 to 1000 ohms. A short circuit here indicates a failure of filter capacitor 13C, power supply transformer 1T, transient suppressor 1BD, or one or more of the field power supply diodes 15D, 16D, and 17D or 18D. (Failure of a field diode indicates a short in the motor field circuit.) Next, inspect the wiring for bare wires, loose terminals, and stray wire strands at wire terminations, and verify that the field circuit and regulating circuit are isolated from building ground (refer to the wiring checks in the "Prepower Checks and Adjustments" section for grounding exceptions). Finally, disconnect one motor field lead and measure the motor field resistance while very slowly rotating the motor shaft through a full 360°. The field resistance should be between 200 and 500 ohms and should be the same regardless of the shaft position.

Motor Hunts

Too much IR compensation may cause some motor-load combinations to hunt. Turn the IR compensation adjustment CCW until the hunting stops.

If adjustments of the IR compensation do not correct the problem, verify that the motor is not running faster than rated speed, and if the control is being used on 50 Hz line voltage, verify that resistor 140R has been cut out of the circuit as shown in Figure 7. Then refer to the section of this book entitled "Prepower Checks and Adjustments" and check the current limit setting. If it was not set per Table II, turn off the line circuit breaker and correct the setting. If it was set per Table II, turn off the power circuit breaker and change the current limit adjustment to the setting appropriate to the next lower impedance group of motors in Table II.

REPAIR

Motor Repair

WARNING

HIGH VOLTAGE ELECTRIC SHOCK CAN CAUSE SERIOUS OR FATAL INJURY. BE CAREFUL NOT TO COME IN CONTACT WITH LIVE PARTS. THE CONTROLLER STOP SWITCH DOES NOT REMOVE VOLTAGE FROM THE MOTOR FIELD. ALWAYS TURN OFF THE LINE CIRCUIT BREAKER BEFORE ATTEMPTING TO SERVICE MOTOR.

The motor can be repaired just as a standard DC motor by any competent motor repair service. For replacement parts or motor service, take the motor model number from the motor nameplate and contact the nearest service shop authorized by the motor manufacturer.

Controller Repairs

Normal field repair should be limited to replacing component boards, options, fuses, circuit breakers, and the SCR's. The complexity of the test sequence required to verify proper operation of a component board after repair makes it inadvisable to attempt field repair of the printed circuit board. Replacement of the complete printed circuit board is the quickest and easiest approach to troubleshooting and repair.

When returning component boards for exchange, pack them carefully to prevent damages from occuring in transit.

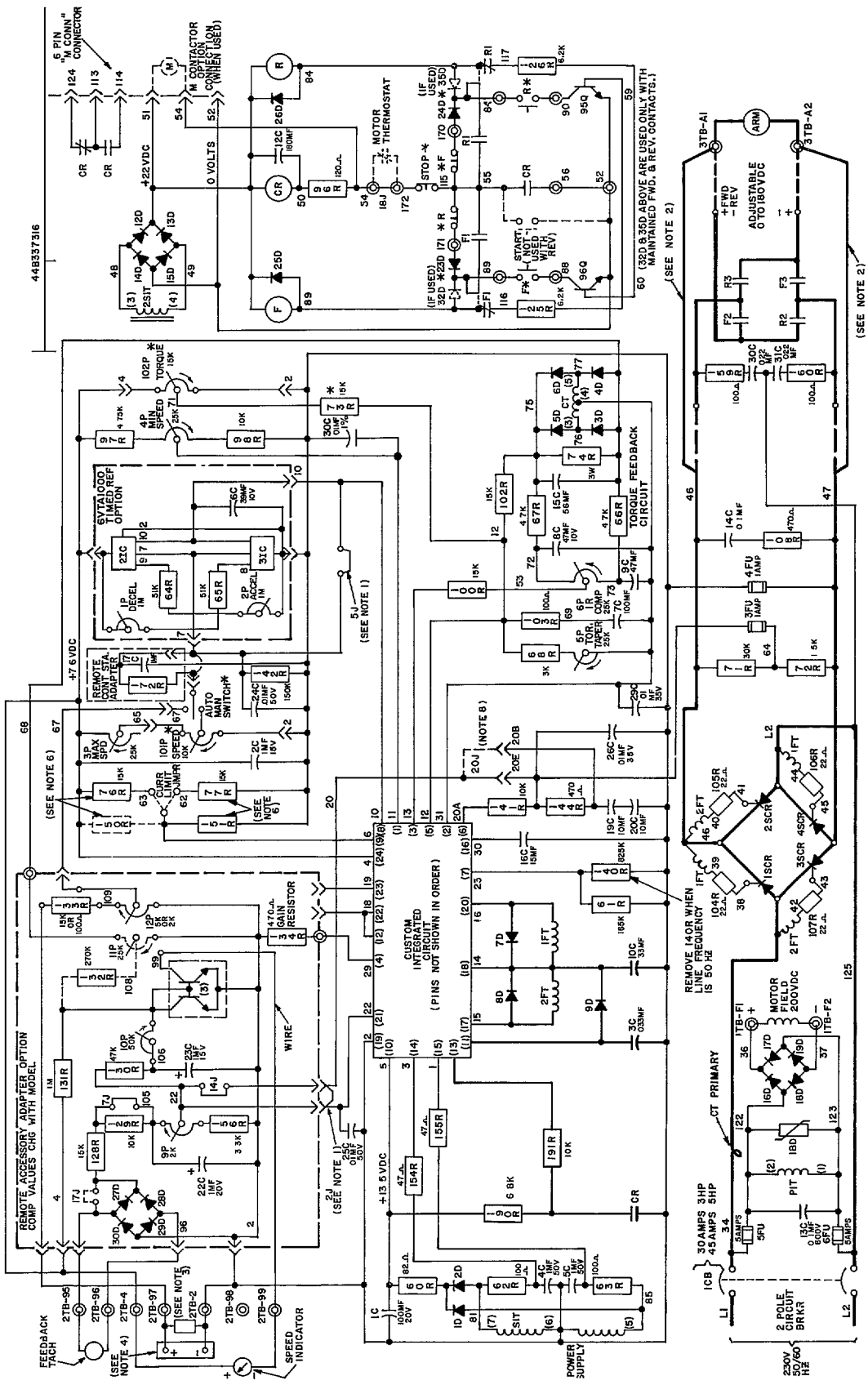


Figure 8. Statotrol II DC Drives, 3 HP and 5 HP, Elementary Diagram

When replacing SCR's, it is recommeded that a thin coating of silicone grease be applied to the replacement SCR mounting threads before installing. A good recommended silicone grease is GE G322L Versilube™, available from General Electric Co.

The recommended maximum torque for the SCR's is 20 inch-pounds for the 3 hp and 100 inch-pounds for the 5 hp. Torqueing the SCR's beyond these limits may damage the SCR's or the mounting threads.

RENEWAL AND SPARE PARTS

Replacement parts can be ordered from the nearest sales office of the General Electric Company, Replacements for control stations and options should be ordered by the "6V" catalog number which appears on the control station or option originally purchased with the control.

REMOTE CONTROL STATION ADAPTERS 6VRC2002 AND 6VRC2003

DESCRIPTION

The 6VRC2002 and 6VRC2003 remote control station adaptors plug into the Statotrol II controller and provide terminals for connection to a control station with maintained "Forward" and "Reverse" switch contacts. The 6VRC2002 is for use with the chasis-mount version of the controller and the 6VRC2003 is for use with the enclosed version of the controller.

INSTALLATION

Refer to Figure 2 in this instruction book for proper mounting locations. The captive screws at the four corners of the remote control station adaptor secure the bracket to the controller.

ELECTRICAL CONNECTIONS

Figure 9 shows connections for a typical remote control station. The relay coil circuit (F, R and override stop switchings) is an isolated, 24V DC, NEC class II circuit. If local codes require this circuit to be grounded, connect ground lead to terminal 51. Do **not** ground at more than one point.

If Auto-Man switch is not used, connect the wiper of the speed control potentiometer directly to terminal 132. If limit switches are not used, connect switches F&R directly to terminal 88 and 145 respectively. If motor thermostat is not used, connect the override stop switch directly to terminal 172. The use of an override stop switch can be avoided by connecting the forward, reverse, stop circuit as shown in Figure 10 or Figure 11.

The Auto-Man switch, and any other switch used in the speed control potentiometer circuit or the torque control potentiometer circuit, must have gold plated contacts suitable for dry circuit operation.

Speed control, torque control, and Auto-Man switch wires must be kept separate from all other wires. **Do not** run these wires in conduits or wire harnesses with power wiring or relay coil wiring. **Do not allow any point in this circuit to contact building ground.** Shielded wire is usually not necessary. However if shielded wire is used for the speed or torque control wiring, connect the shield to circuit 2 at the controller terminal board. All AC brakes and relays should be suppressed.

CAUTION

IF SHIELDED WIRE IS USED, THE SHIELD MUST BE INSULATED. ALLOWING THE SHIELD TO CONTACT BUILDING GROUND WILL CAUSE DAMAGE TO THE CONTROLLER.

CAUTION

NEVER ATTEMPT TO REVERSE WHILE MOTOR IS RUNNING OVERSPEED. USE MAXIMUM SPEED ADJUSTMENT ON CONTROLLER MAIN COMPONENT BOARD TO LIMIT MOTOR SPEED.

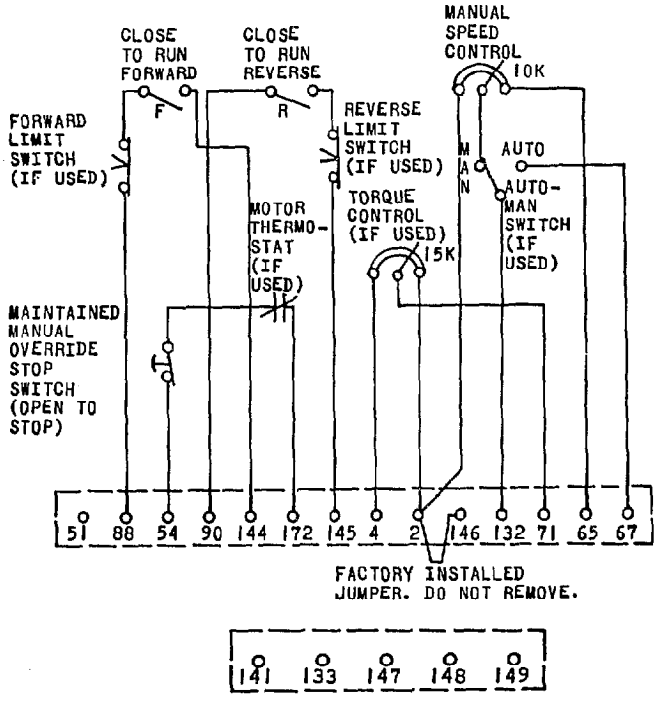
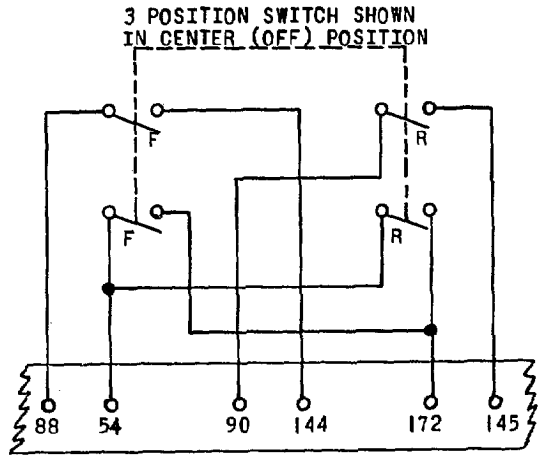


Figure 9. Typical Remote Control Station Connection Diagram



NOTE: Close F to run forward, Close R to run reverse.

Figure 11. Partial Connection Diagram Showing Connection for 4 Pole 3 Position Forward, Reverse, Off Switch

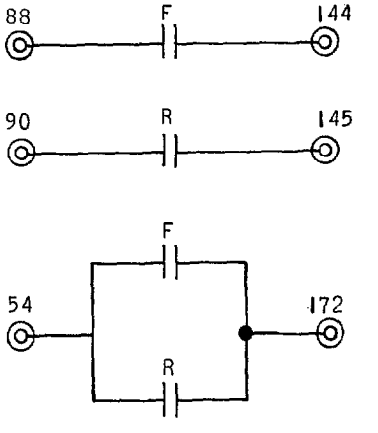


Figure 10. Connections for Forward, Reverse, Stop Switching by Pair of 2 Pole Relays

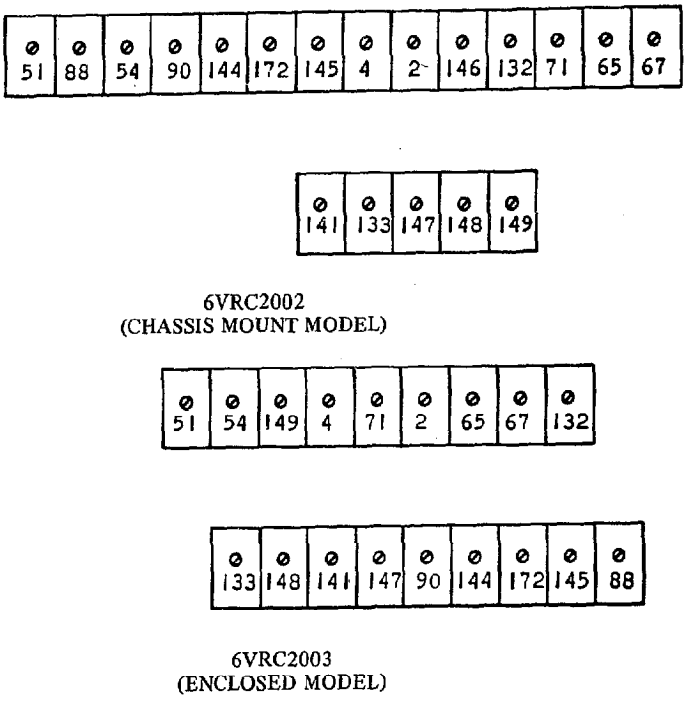


Figure 12. Terminal Arrangements on Open Chassis Mount Model and Enclosed Model

REMOTE ACCESSORY ADAPTER

6VFB1000 & 1001 AND 6VFB1100 & 1101

DESCRIPTION

The 6VFB1000 and 1001 adapters are small printed circuit boards which plug into the main component board of Statotrol II controllers. The adapter option interfaces between the control circuit and the proper remote accessories to provide tachometer feedback, remote voltage reference following (tachometer follower applications), and speed indication.

The 6VFB1100 and 1101 options are similar to the above units, except that the options are designed to interface with remote signal generating instruments (instrument follower applications).

NOTE

When any 6VFB series option is installed, a feedback tachometer **must** be used. The follower and speed indication features may either be used or ignored.

INSTALLATION

Refer to Figure 2 in this instruction book for proper location and orientation for installation. The adapter is mounted and connected by two screws and a row of eleven pins and receptacles. Two of the receptacles on the main component board are connected together by a factory installed metal jumper (terminals 20 and 22). This jumper must be removed before the adapter option is installed. To install the option, position it carefully so that each pin is properly mated with its receptacle, press the option board down toward the main component board, and then insert and tighten the two screws. The screws provide electrical connection in addition to providing physical mounting.

CAUTION

SCREWS OF THE PROPER LENGTH ARE SUPPLIED WITH THE OPTION. DO NOT SUBSTITUTE LONGER SCREWS, SINCE LONGER SCREWS MAY SHORT THE CONTROL CIRCUIT TO THE CHASSIS.

Note: The wire attached to the remote accessory adapter unit should be connected to terminal 99 on terminal strip 2TB.

ELECTRICAL CONNECTIONS

Connection diagrams and wiring instructions are provided in the controller instructions under the heading "Remote Accessory Wiring", and on Figure 13 on the following page

To avoid electrical noise pickup, it is necessary to keep the remote accessories wiring separate from all other wiring. Do not run these wires through conduits with power conductors or relay coil wiring.

If separation of the remote accessories wiring is physically impossible or impractical to do, then insulated, shielded cabling **must** used. Connect the shield to circuit 2 at the controller 2TB terminal board. Do not connect the shield at the other end but **do** insulate it so it cannot come in contact with other circuits or building ground. All AC brakes and relays should be suppressed.

CAUTION

IF SHIELDED WIRE IS USED, THE SHIELD MUST BE INSULATED. ALLOWING THE SHIELD TO CONTACT BUILDING GROUND WILL CAUSE DAMAGE TO THE CONTROLLER.

CAUTION

ALL REMOTE CIRCUITRY CONNECTED TO THE STATOTROL II DRIVE MUST BE COMPLETELY ISOLATED FROM GROUND, OR A LINE ISOLATION TRANSFORMER MUST BE USED TO SUPPLY POWER TO THE CONTROLLER. IN NO CASE SHOULD MORE THAN ONE CIRCUIT POINT BE GROUNDED. IF MORE THAN ONE DRIVE IS TO FOLLOW THE SAME SIGNAL, EACH DRIVE MUST A LINE ISOLATION TRANSFORMER.

VOLTGE LIMITATIONS

CAUTION

OVERVOLTAGE FROM A FEEDBACK TACHOMETER OR A REMOTE REFERENCE SIGNAL MAY DAMAGE THE CONTROL.

TABLE IV

Option	Feedback Voltages				Follower (remote reference) voltage	
	Max** Overvoltage		Nominal for** Rated Speed		Max Overvoltage	Nominal for Rated Speed
	w/7J	w/o7J	w/7J	w/o7J		
6VFB1000	50	75	50	50	50	35
6VFB1001	100	125	70	87 5	100	70
6VFB1100	50	75	35	50	12.5*	8.0
6VFB1101	100	125	70	87 5	12 5*	8.0

*When connected to a current source a protective shunt resistor is required. Refer to Figure 6 in Controller instruction book
When connected to a voltage source, a protective series resistor is required. Refer to Figure 13 on this sheet.

**Jumper 7J is factory installed on the option Remove 7J by cutting it out to increase maximum and nominal feedback voltage as shown.

ADJUSTMENTS

WARNING

HIGH VOLTAGE ELECTRIC SHOCK CAN CAUSE SERIOUS OR FATAL INJURY. BE CAREFUL NOT TO COME IN CONTACT WITH LIVE PARTS. LINE VOLTAGE IS EXPOSED WHEN THE STATOTROL II COVER IS REMOVED. ELECTRICAL POWER SHOULD BE DISCONNECTED AT THE BRANCH CIRCUIT BREAKER BEFORE ANY WORK OTHER THAN POTENTIOMETER ADJUSTMENTS IS PERFORMED INSIDE THE CONTROLLER. INTERNAL ADJUSTMENTS SHOULD BE PERFORMED BY QUALIFIED ELECTRICIANS.

FEEDBACK SIGNAL ADJUSTMENT (If Instrument Follower Feature is not Used)

The maximum speed adjustment, potentiometer 3P, on the main controller component board does not, by itself, limit the maximum motor speed when tachometer feedback is used. However, to get the correct signal level for proper operation, the maximum speed adjustment should be left at its factory setting. If the factory setting has been disturbed, turn 3P to approximtely 80% of its possible CW rotation.

After all items in the 'Prepower Checks and Adjustments" section of the controller instruction book have been completed, set the control to "manual" (if the control station has "auto" and "manual" settings), start the control (in either forward or reverse) and turn the speed control knob to full speed. Rotate 9P on the adapter option until the motor runs at the highest speed needed for the particular application. It is recommended that maximum operating speed be limited to the nameplate rating of the motor since the drive may not perform as specified if the motor is in an overspeed condition.

TACHOMETER FOLLOWER ADJUSTMENT WITH FEEDBACK PREVIOUSLY ADJUSTED

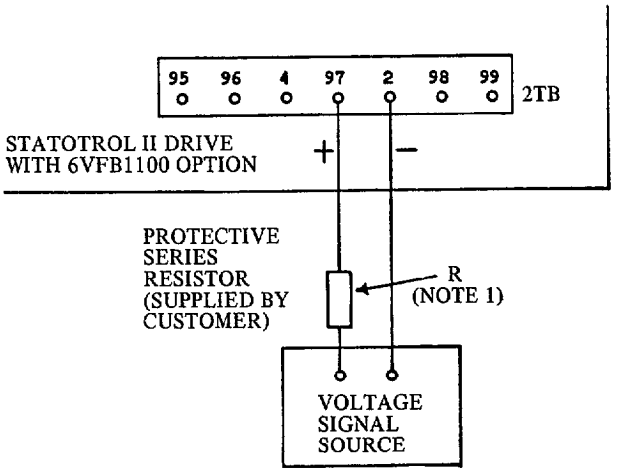
The tachometer follower feature on the 6VFB1000 & 1001 should be adjusted after the tachometer feedback feature has been adjusted with the control in "manual".

To adjust the follower feature, set the tachometer to the output which calls for the highest Statotrol motor operating speed required by the application. Press the "auto" and "start" buttons on the control station, and adjusting potentiometer 12P on the remote accessory adapter option until the Statotrol motor runs at the correct speed (do not exceed rated speed).

ADJUSTMENT WHEN INSTRUMENT FOLLOWER FEATURE AND TACHOMETER FEEDBACK ARE BOTH USED

After all items in the "Prepower Checks and Adjustments" section of the controller instruction book have been completed, adjust the follower and feedback adjustments as follows:

1. Set the output of the reference generator to the signal which must correspond to zero speed of the Statotrol motor. Press the "start" and "auto" buttons on the control station, and turn the minimum speed adjustment potentiometer, 4P, clockwise until the Statotrol motor starts to turn. Then, slowly turn 4 P CCW until the motor just stops.



NOTE 1: R = 250•V − 2000 OHMS, WHERE V IS THE MAXIMUM OUTPUT VOLTAGE OF THE VOLTAGE SIGNAL SOURCE. IF V IS LESS THAN 8 VOLTS, CONTACT THE FACTORY FOR SPECIAL INSTRUCTIONS.

Figure 13. Connection Diagram for Remote Voltage Signal Source

2. Press the "Manual" button on the Statotrol control station, and turn the speed control knob to full speed. Locate 9P on the remote accessory adapter option, and turn 9P until the motor runs at the highest speed required for the application (do not exceed rated speed).

3. Press the "Auto" button on the control station, and set the output of the current generator to the current which corresponds to the highest motor speed required by the application.

CAUTION

THE CORRECT PROTECTIVE RESISTOR MUST BE CONNECTED AS SHOWN IN THE REMOTE ACCESSORY CONNECTION DIAGRAM OF THE CONTROLLER INSTRUCTION BOOK FOR CURRENT SIGNAL INSTRUMENTS OR AS SHOWN IN FIGURE 13 FOR VOLTAGE SIGNAL INSTRUMENTS.

Locate 12P on the remote accessory adapter option and rotate 12P until the motor runs at the correct speed. It may be necessary to "fine tune" the drive by repeating steps 1, 2, and 3.

SPEED INDICATION ADJUSTMENT

The speed indicator should be adjusted after the tachometer feedback and follower adjustments are complete Start the motor and set the speed control knob near full speed. Measure the speed of the motor with a strobe or a tachometer, and adjust-potentiometer 10P on the remote accessory adapter option until the speed meter indicates the actual motor speed.

VOLTAGE SIGNAL FOLLOWER

In cases where it is necessary to follow a DC voltage signal from an instrument, the 6VFB1100 option may be used with a voltage generator connected as shown in Figure 13. The voltage applied across 2TB-97 and 2TB-2 must be of the indicated polarity and a protective series resistor must be used to limit the applied voltage to 12 volts maximum during overvoltage conditions.

Adjustments should be performed as described on this sheet under the heading "Adjustment when Instrument Follower and Tachometer Feedback are Both Used".

REMOTE ACCESSORY ADAPTER 6VTF1000, 1001, 1002 AND 6VMA1000

DESCRIPTION

The 6VTF series adapter option is a small printed circuit board which plugs into the main component board of Statotrol II controllers. The adapter option interfaces between the control circuit and the proper remote accessories to provide remote voltage reference following (tachometer follower applications), and speed indication.

The 6VMA1000 option is similar to the 6VTF series except that the 6VMA1000 option is designed to interface with remote reference signal generating instruments (instrument follower applications).

The follower feature and the speed indication feature may both be used at the same time, or one can be used and the other ignored.

INSTALLATION

Refer to Figure 2 in this instruction book for proper location and orientation for installation. The adapter is mounted and connected by two screws and a row of eleven pins and receptacles. Two of the receptacles on the main component board are connected together by a factory installed metal jumper (terminals 20 and 22). This jumper must be removed before the adapter option is installed. To install the option, position it carefully so that each pin is properly mated with its receptacle, press the option board down toward the main component board, and then insert and tighten the two screws. The screws provide electrical connection in addition to providing physical mounting.

CAUTION

SCREWS OF THE PROPER LENGTH ARE SUPPLIED WITH THE OPTION. DO NOT SUBSTITUTE LONGER SCREWS, SINCE LONGER SCREWS MAY SHORT THE CONTROL CIRCUIT TO THE CHASSIS.

NOTE: The wire attached to the remote accessory adapter unit should be connected to terminal 99 on terminal strip 2TB.

ELECTRICAL CONNECTIONS

Connection diagrams and wiring instructions are provided in the controller instruction book under the heading "Remote Accessory Wiring" and on Figure 14.

To avoid electrical noise pickup, it is necessary to keep the remote accessories wiring separate from all wiring. Do not run these wires through conduits with power conductors or relay coil wiring.

If separation of the remote accessories wiring is physically impossible or impractical to do, then insulated, shielded cabling must be used. Connect the shield to circuit 2 at the controller 2TB terminal board. Do not connect the shield at the other end but do insulate it so it cannot come in contact with other circuits or building ground. All AC brakes and relays should be suppressed.

CAUTION

IF SHIELDED WIRE IS USED, THE SHIELD MUST BE INSULATED. ALLOWING THE SHIELD TO CONTACT BUILDING GROUND WILL CAUSE DAMAGE TO THE CONTROLLER.

CAUTION

ALL REMOTE CIRCUITRY CONNECTED TO THE STATOTROL II DRIVE MUST BE COMPLETELY ISOLATED FROM GROUND, OR A LINE ISOLATION TRANSFORMER MUST BE USED TO SUPPLY POWER TO THE CONTROLLER. IN NO CASE SHOULD MORE THAN ONE CIRCUIT POINT BE GROUNDED. IF MORE THAN ONE DRIVE IS TO FOLLOW THE SAME SIGNAL, EACH DRIVE MUST HAVE A SEPARATE LINE ISOLATION TRANSFORMER.

CAUTION

OVERVOLTAGE FROM A REMOTE SIGNAL GENERATOR (EITHER A TACHOMETER OR AN INSTRUMENT) MAY DAMAGE THE CONTROL.

TABLE V

OPTION	MAXIMUM ALLOWABLE APPLIED VOLTAGE	NOMINAL VOLTAGE FOR FULL SPEED
6VTF1000	50V	35VDC
6VTF1001	70V	50VDC
6VTF1002	110V	70VDC

When the 6VMA1000 option is used with a current signal generator, the proper protective shunt resistor (see the remote accessory connection diagram in the controller instruction book) must be connected between terminals 97 and 2 on 2TB)

ADJUSTMENTS

WARNING

HIGH VOLTAGE ELECTRIC SHOCK CAN CAUSE SERIOUS OR FATAL INJURY. BE CAREFUL NOT TO COME IN CONTACT WITH LIVE PARTS. LINE VOLTAGE IS EXPOSED WHEN THE STATOTROL II COVER IS REMOVED. ELECTRICAL POWER SHOULD BE DISCONNECTED AT THE BRANCH CIRCUIT BREAKER BEFORE ANY WORK OTHER THAN POTENTIOMETER ADJUSTMENT IS PERFORMED INSIDE THE CONTROLLER. INTERNAL ADJUSTMENTS SHOULD BE PERFORMED BY QUALIFIED ELECTRICIANS.

TACHOMETER FOLLOWER ADJUSTMENT OF 6VTF1000, 1001, 1002

After all items in the "Prepower Checks and Adjustments" section of the controller instruction book have been completed, set the minimum speed adjustment as described in the controller instruction book, and then set the tachometer output to the level which corresponds to the highest normal operating speed of the Statotrol motor as required by the application. Press the "Auto" and "Start" buttons on the control station, and adjust potentiometer 12P on the remote accessory adapter option until the Statotrol motor runs at the correct speed. Do not exceed rated speed.

INSTRUMENT FOLLOWER ADJUSTMENT OF 6VMA1000

First complete all items in the "Prepower Checks and Adjustments" section of the controller instruction book. Then adjust the instrument follower feature by the following sequence.

1. Set the output of the reference generator to the level which must correspond to zero speed of the Statotrol motor, press the "start" and "auto" buttons on the control station, and turn the minimum speed adjust potentiometer, 4P, CW until the motor starts, and then 4P CCW until the motor just stops.

2. Set the output of the reference generator to the signal level which corresponds to the highest motor speed required by the application and turn 12P on the remote

accessory adapter option until the motor runs the correct speed. Do not exceed rated speed. It may be necessary to "fine tune" the drive by repeating steps 1 and 2.

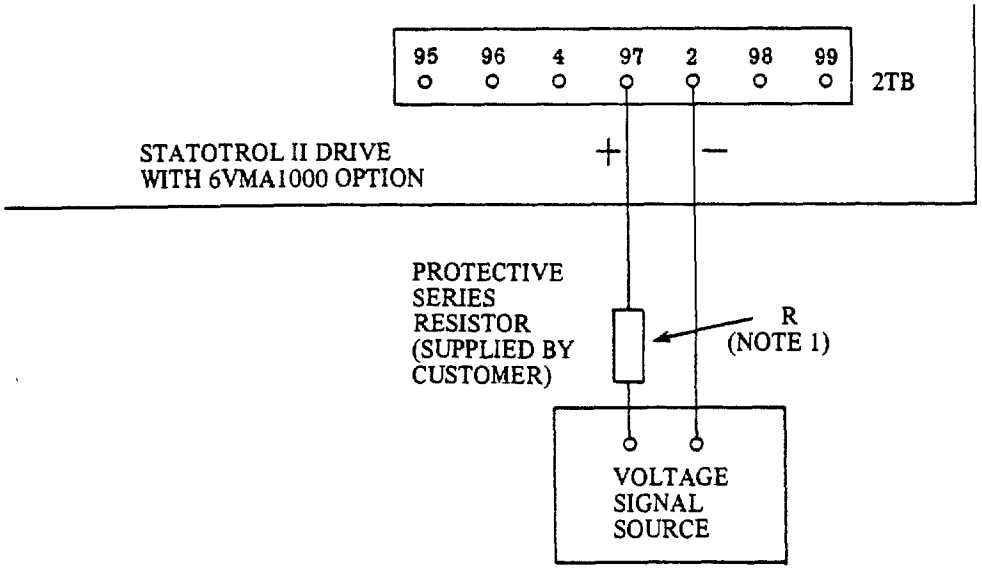
SPEED INDICATION ADJUSTMENT

After all items in the "Prepower Checks and Adjustments" section of the controller instruction book have been completed, run the motor at no load and near rated speed. Measure the speed of the motor with a strobe or a tachometer, and adjust 10P on the remote accessory adapter option until the speed indicator reading is correct. Then load the motor, measure the speed with strobe or tachometer again, and adjust potentiometer 11P on the remote accessory adapter option until the speed indicator reading is correct.

DC VOLTAGE SIGNAL FOLLOWER

In cases where it is necessary to follow a DC voltage signal, from an instrument, the 6VMA1000 option may be used with a voltage generator connected as shown. The voltage applied across 2TB-97 and 2TB2 must be of the indicated polarity and a protective series resistor must be used to limit the applied voltage to 12 volts maximum during overvoltage connection.

Adjustment should be performed as described on this sheet under the heading "Instrument Follower Adjustment of 6VMA100".



NOTE 1: R = (250) (V) - 2000 OHMS, WHERE V IS THE MAXIMUM OUTPUT VOLTAGE OF THE VOLTAGE SIGNAL SOURCE. IF V IS LESS THAN 8 VOLTS, CONTACT THE FACTORY FOR SPECIAL INSTRUCTIONS.

Figure 14. Connection Diagram for Remote Voltage Signal Source

TIMED ACCELERATION AND DECELERATION OPTION 6VTA1000, 6VTA1001, AND 6VTA1002

DESCRIPTION

The Statotrol II timed acceleration and deceleration option consists of several components on a small printed circuit board which plugs into the main controller component board. This option provides adjustable linear motor acceleration in response to a rapid change in the reference signal.

The 6VTA1000 option is adjustable from 2 seconds to 30 seconds. The 6VTA1001 option is adjustable from 5 seconds to 60 seconds. The 6VTA1002 is adjustable from 0.75 seconds to 8 seconds. Only the 6VTA1002 is suitable for use with the Statotrol II Regenerative Braking Option.

INSTALLATION

Refer to Figure 2 in this instruction book for proper mounting orientation and location. The timed acceleration and deceleration option has a plastic standoff for mounting

strength and 4 pins for electrical connection to receptacles on the main component board. Two of the receptacles are connected together by a factory installed jumper. This jumper must be removed before the option is installed. When installing the option, use care to avoid bending the connection pins. Be certain that each pin mates properly with its receptacle. Press the option board down firmly until the plastic mounting standoff snaps into place.

OPERATION AND ADJUSTMENT

When the "start" button is pressed, motor speed will immediately come up to the speed set by the "minimum speed" potentiometer and then accelerate to set speed at a rate determined by the position of the "acceleration rate" potentiometer. If the "minimum speed" potentiometer is set counterclockwise from its center position, there will be a delay between "start button" actuation and motor start up.

To increase the rated acceleration from minimum speed to set speed, turn the "acceleration rate" potentiometer clockwise.

When the motor is running at a steady speed and the reference is quickly turned to zero, the motor will decelerate to the speed set by the "minimum speed" potentiometer at a rate determined by the setting of the "deceleration rate" potentiometer. To increase the rate at which the motor slows down, turn the "deceleration rate" potentiometer clockwise.

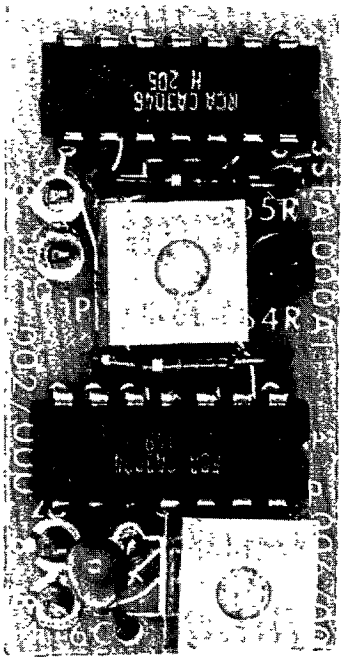


Figure 15. 6VTA . . . . . Timed Acceleration & Deceleration Option

M CONTACTOR AND BRAKE CONTACTOR OPTIONS 6VMC1000 AND 6VMC1001

DESCRIPTION

6VMC1000

The M contactor option provides contacts which open the motor armature circuit when the stop button is pressed. The CR relay on the standard controller component board disconnects the SCR gates, so the M contactor option is necessary only in applications which require mechanical contacts in the armature circuit for added confidence that no electrical power can reach the motor while it is stopped

6VMC1001

This option is primarily intended for switching a motor mounted brake or clutch, but can be used for many other purposes. The user must make his electrical connections at the relay terminals. The 6VMC1001 provides a form C (SPDT) contact rated 30 amps resistive at 277V AC. When properly connected, the contactor is energized while the drive is running. The 6VMC1000 option may be converted to a 6VMC1001 option by removing and discarding the wire which is factory connected to a contact terminal on the 6VMC1000 option.

INSTALLATION

Refer to Figure 2 in this instruction book for the mounting location of the 6VMC1000 option. The 6VMC1001 option mounts in the same position. Hardware is provided for mounting the option on the controller.

ELECTRICAL CONNECTIONS

Relay Coil Connections for 6VMC1000 & 6VMC1001

Two wires from the contactor coil are terminated in a connector. This connector must be mated with the male connector marked "M" CONN." and located on the controller main component board.

Power Circuit Connections for 6VMC1000

The armature circuit must be connected as shown below. To accomplish this, remove wire number 46 from 3TB-A1 and connect wire number 46 to the contactor as shown. Then locate wire number 92 which is factory connected to the contactor and connect the free end of wire number 92 to 3TB-A1.

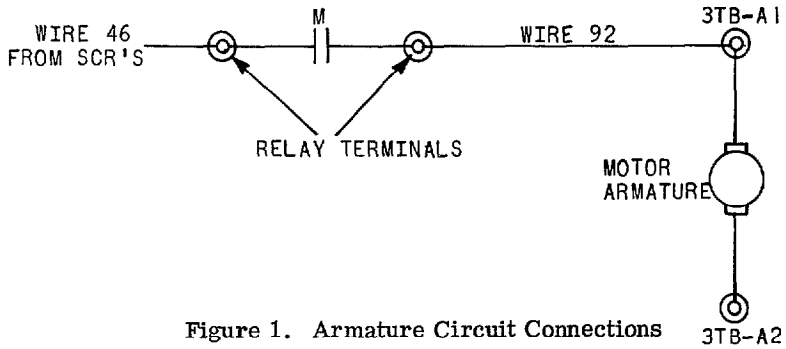


Figure 1. Armature Circuit Connections

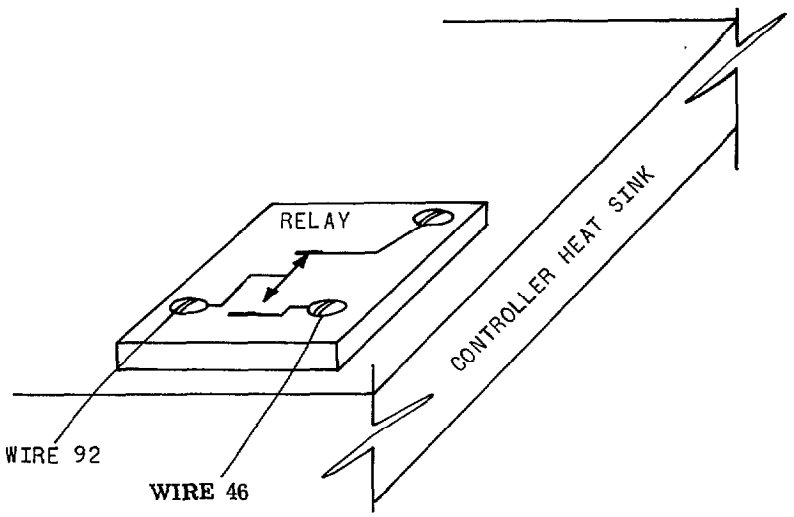


Figure 16. Contactor Terminal Arrangement

REVERSING OPTION
6VRK1001

DESCRIPTION

The 6VRK1001 reversing option consists of several small components mounted on a bracket with two prewired contactors. With this option properly installed, the Statotrol II controller provides controlled torque plug reversing for DC motors rated 3 HP and 5 HP.

INSTALLATION

For proper mounting location and orientation, refer to Figure 2 in this instruction book. Hardware for securing the bracket to the controller chassis is provided with the option.

ELECTRICAL CONNECTIONS

Relay Coil Connections

The relay coil wiring is terminated in a connector. This connector must be mated with the male connector, marked "REV. CONN." located on the controller main component board.

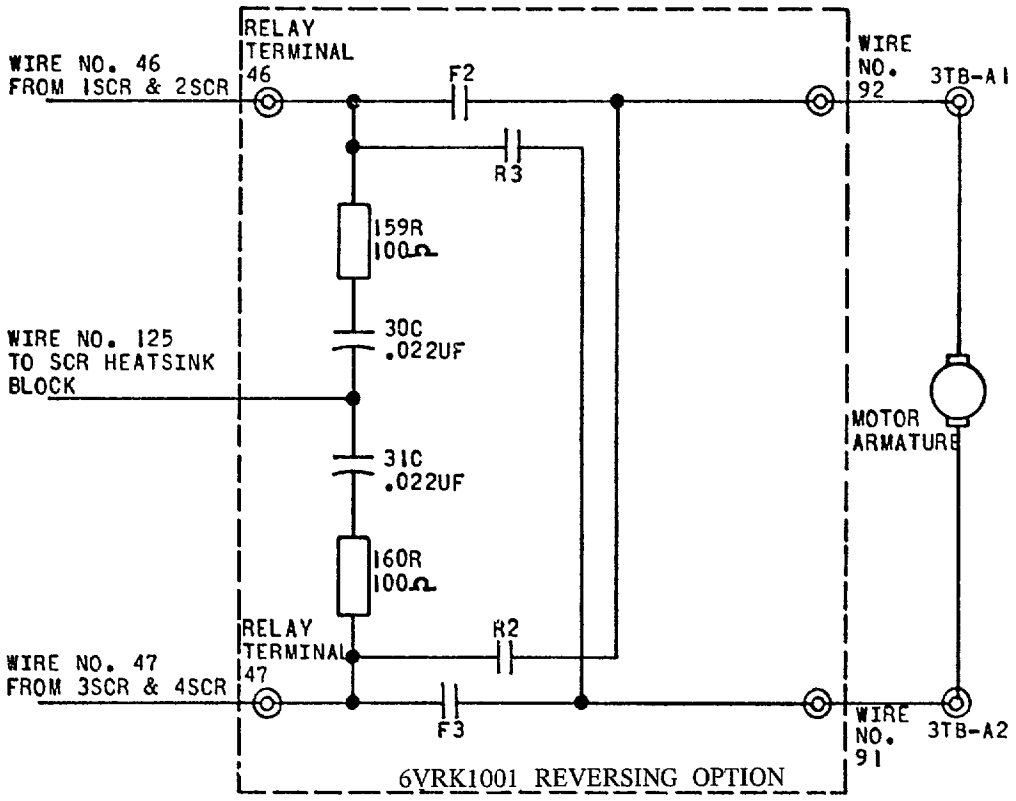


Figure 17. Motor Armature Circuit with 6VRK1001 Reversing Option Installed

Power Circuit Connections

The motor armature circuit must be connected as shown below. Refer to Page 27 (Figure 18) for terminal locations on the contactors. After the option has been mounted, proceed with the following connection sequence.

1. Locate wire number 46 which runs from the SCR's to terminal 3TB-A1. Note: On 3 hp units, wire 46 runs from the SCR's to the insulated standoff and then to 3TB-A1. Disconnect this wire from 3TB-A1 and connect it to contactor terminal 46 on the reversing option (see Page 27).

2. Locate wire number 47 which runs from the SCR's to terminal 3TB-A2. Disconnect this wire from 3TB-A2 and connect it to contactor terminal 47 on the reversing option (see Page 27).

3. Locate wire number 92 (AWG. 12) which is factory connected to the reversing option. Connect the free end of wire number 92 to terminal 3TB-A1.

4. Locate wire number 91 (AWG. 12) which is factory connected to the reversing option. Connect the free end of wire number 91 to terminal 3TB-A2.

5. Locate wire number 125 (AWG. 18) which is factory connected to the reversing option. Connect the free end of wire number 125 to terminal 125 on the SCR heat sink block as shown in Figure 18.

When the reversing option is properly connected and when motor leads and controller terminals are connected A1 to A1, A2 to A2, F1 to F1, and F2 to F2, the motor will rotate CCW as viewed from the opposite shaft end when the control station button marked "FWD" is pressed.

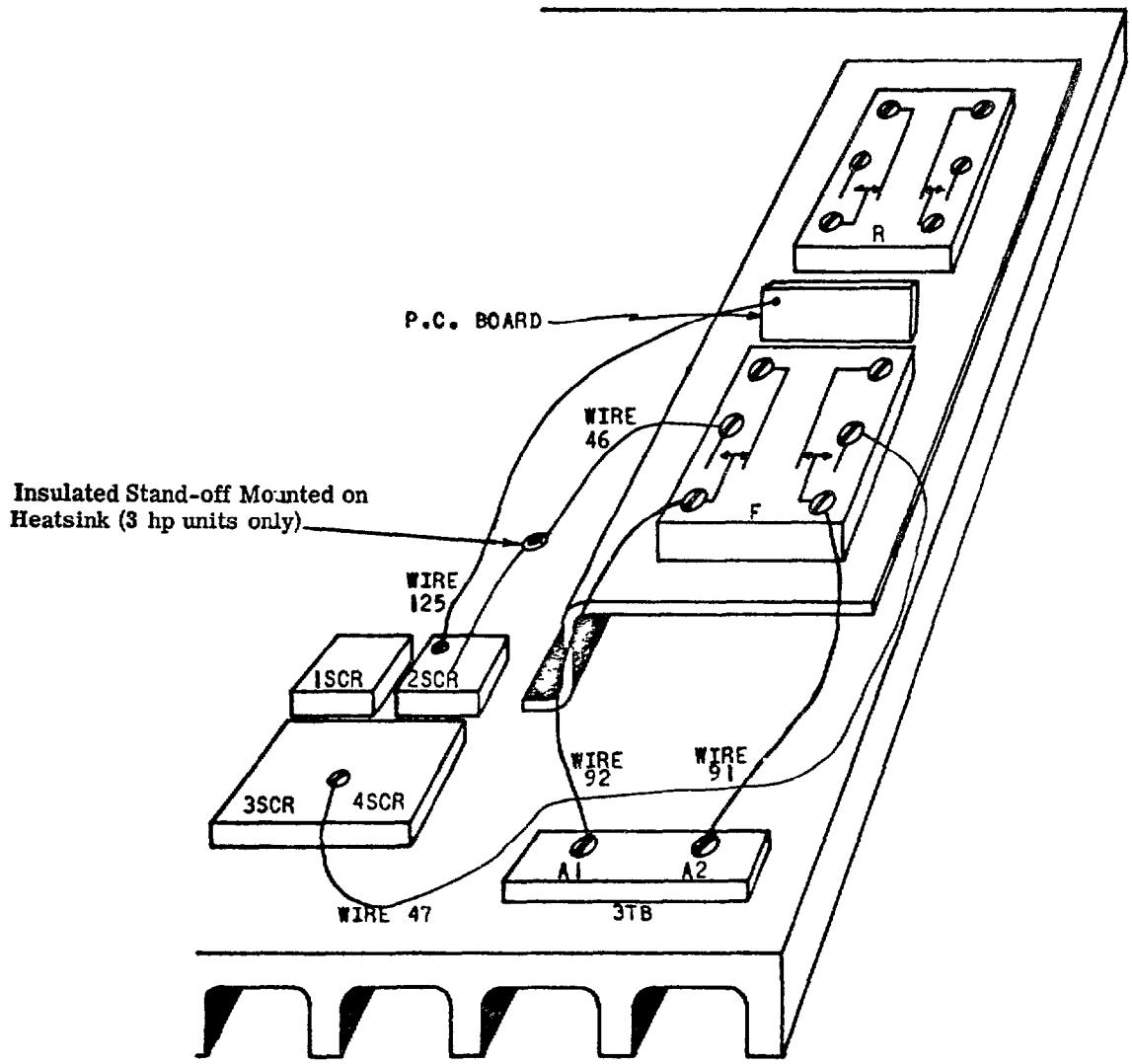


Figure 18. Customer Connection Diagram for 6VRK1001 Reversing Option

REVERSING OPTION WITH DYNAMIC BRAKING 6VDBR2300 AND 6VDBR2500

DESCRIPTION

This option consists of a small component board mounted on a bracket with 3 prewired contactors and a power resistor. With this option properly installed, the Statotrol II controller provides controlled torque plug reversing and dynamic braking. The 6VDBR2300 is for use on 3 HP drives and the 6VDBR2500 is for use on 5 HP drives.

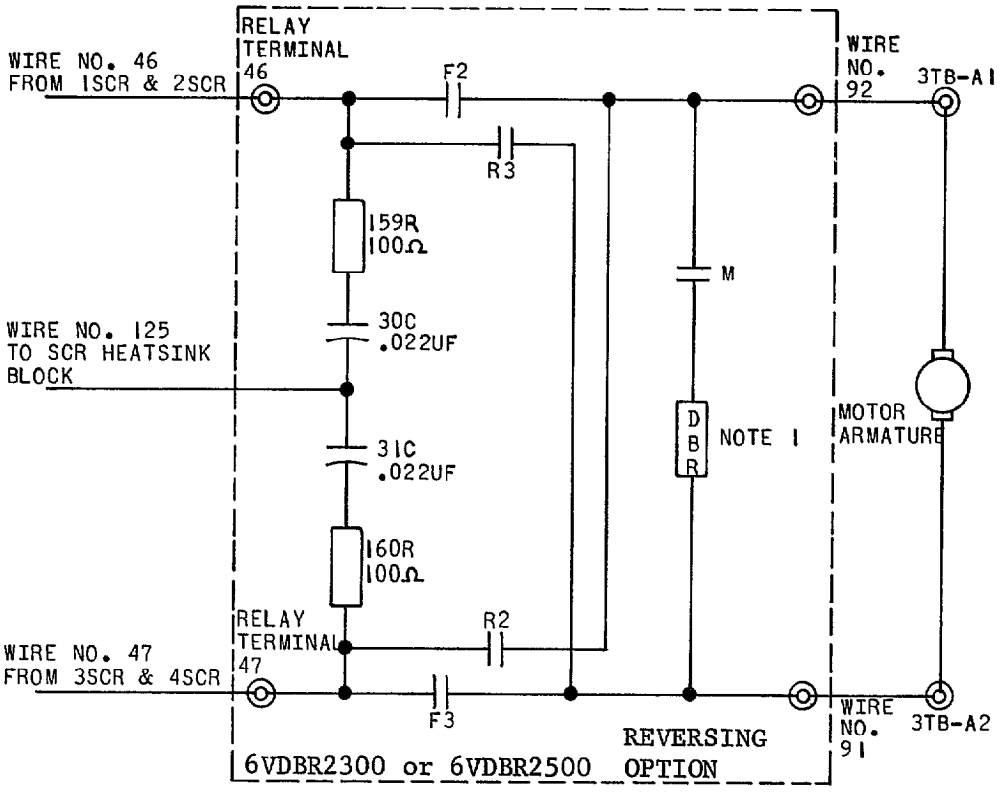
INSTALLATION

For proper mounting location and orientation, refer to Figure 2 in this instruction book. Hardware for securing the bracket to the controller, chassis is provided with the option.

ELECTRICAL CONNECTIONS

Relay Coil Connections

The relay coil wiring is terminted in 2 connectors. The connector marked "REV" must be mated with the male connector marked "REV. CONN." and located on the controller main component board. The connector marked



NOTE 1: THE DYNAMIC BRAKING RESISTOR VALUE CHANGES WITH HORSEPOWER.

Figure 19. Motor Armature Circuit with Option Installed

"M" must be mated with the male connector marked "M CONN." and located on the controller main component board.

Power Circuit Connections

The motor armature circuit must be connected as shown. Refer to Page 26 for terminal locations on the contactors. After the option has been mounted, proceed with the following connection sequence.

1. Locate wire number 46 (AWG 12) which runs from the SCR's to terminal 3TB-A1. NOTE: ON 3 HP UNITS, WIRE 46 RUNS FROM THE SCR's TO THE INSULATED STAND-OFF AND THEN TO 3TB-A1. Disconnect this wire from 3TB-A1 and connect it to contactor terminal 46 on the reversing option (see Figure 20).

2. Locate wire number 47 (AWG 12) which runs from the SCR's to terminal 3TB-A2. Disconnect this wire from 3TB-A2 and connect it to contactor terminal 47 on the reversing option (see Figure 20).

3. Locate wire number 92 (AWG 12) which is factory connected to the reversing option. Connect the free end of wire number 92 on terminal 3TB-A1.

4. Locate wire number 91 (AWG 12) which is factory connected to the reversing option. Connect the free end of the wire number 91 to terminal 3TB-A2.

5. Locate wire number 125 (AWG 18) which is factory connected to the reversing option. Connect the free end of wire number 125 to terminal 125 on the SCR heat sink block as shown.

When the reversing option is properly connected and when motor leads and controller terminals are connected A1 to A1, A2 to A2, F1 to F1, and F2 to F2, the motor will rotate CCW as viewed from the opposite shaft end when the control station button marked "FWD" is pressed.

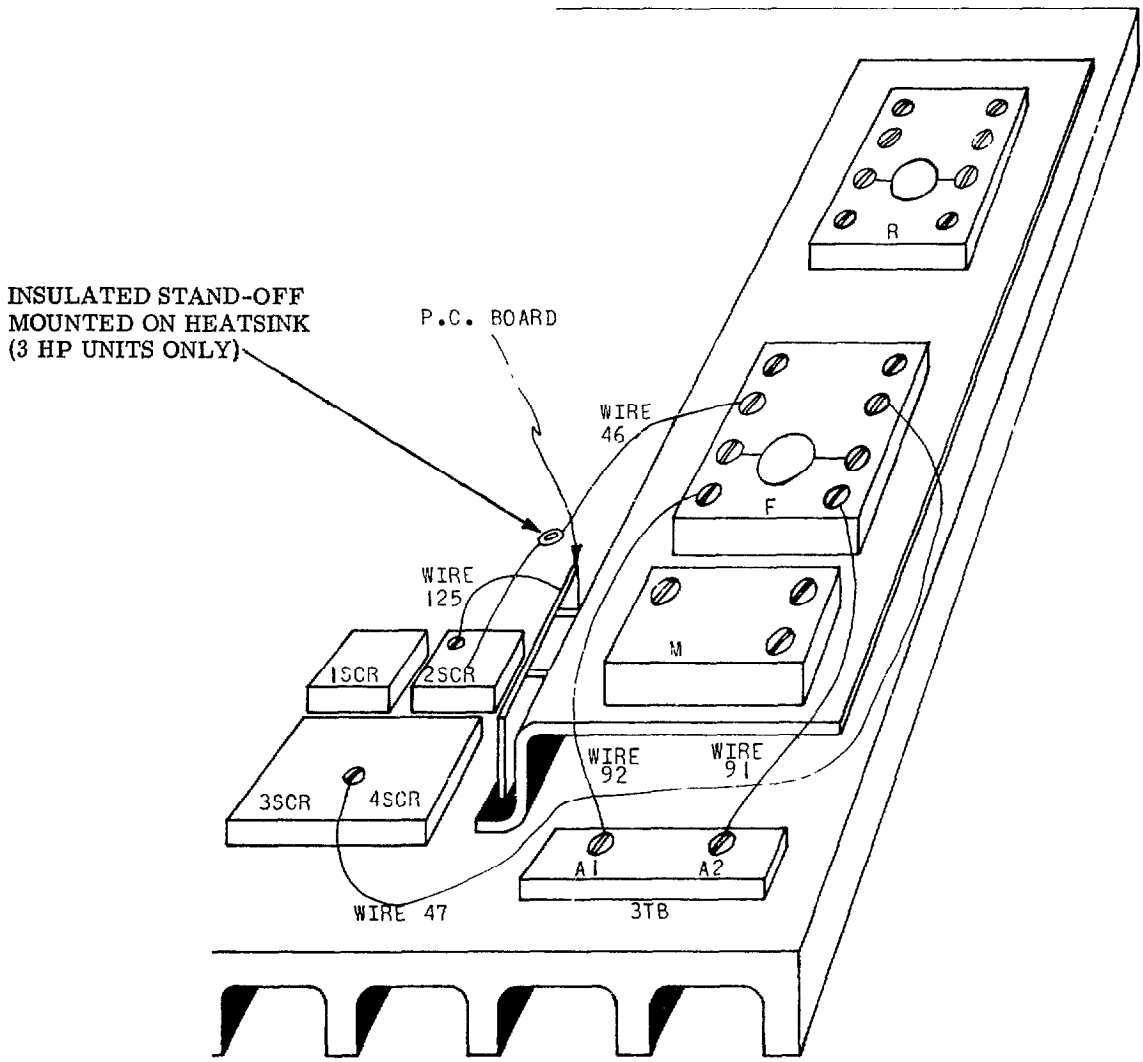


Figure 20. Customer Connection Diagram for 6VDBR2300 and 6VDBR2500 Reversing Options

DYNAMIC BRAKING OPTION 6VDBM2300 AND 6VDBM2500

DESCRIPTION

The dynamic braking option consists of a contactor and a resistor mounted on a bracket. The 6VDBM2300 option is for use with 3 HP motors and the 6VDBM2500 is for use with 5 HP motors.

Normally closed contacts on the contactor connect the resistor across the motor armature whenever the control station stop button is pressed. The motor then acts as a generator and dissipates its rotational energy as heat in the resistor.

INSTALLATION

Refer to Figure 2 in this instruction book for the correct location and orientation for mounting the bracket. Two number 10-32 screws for mounting the bracket to the controller are provided with the option.

ELECTRICAL CONNECTIONS

Two wires from the contactor coil are terminated in a connector. This connector must be mated with the male connector marked "M CONN." and located on the controller main component board.

The power circuit must be connected as shown below, to accomplish this, remove wire number 46 from terminal 3TB-A1 and connect wire number 46 to the contactor as shown below. Then locate wire 92 which is factory connected to the contactor and connect the free end of wire number 92 to terminal 3TB-A1. Then locate wire number 47 which is factory connected to the dynamic braking resistor and connect wire number 47 to terminal 3TB-A2.

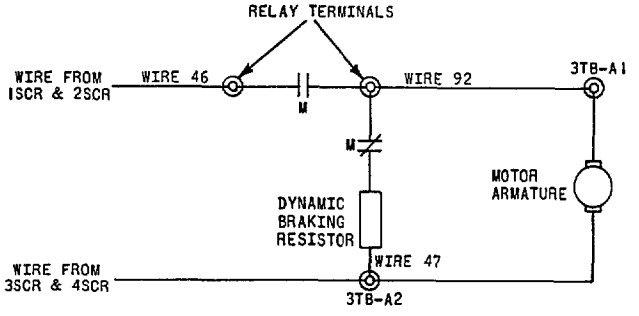


Figure 21. DC Power Circuit Elementary Diagram

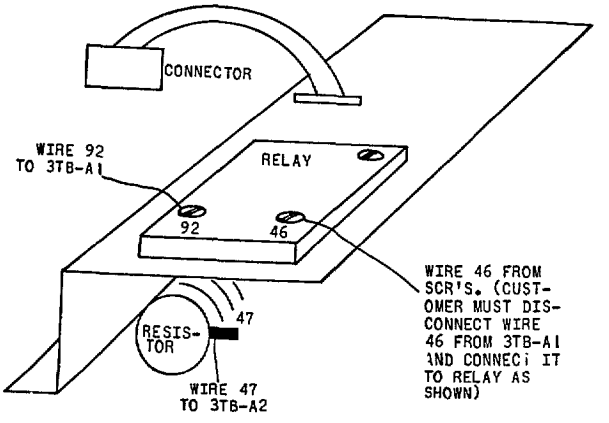


Figure 22. Dyanmic Braking Option Terminal Arrangement

RENEWAL PARTS LIST
FOR 6VFWB, 6VFWC SERIES
3 HP AND 5 HP STATOTROL II CONTROLLERS

DESCRIPTION OF PART OR ASSEMBLY	CATALOG NUMBER	MODEL NUMBER AND HORSEPOWER	
		6VFW2300 3 HP	6VFW2500 5 HP
		QTY. PER UNIT	QTY. PER UNIT
Main Printed Circuit Board	**44B337316G101**	1	
Main Printed Circuit Board	**44B337316G110**		1
Circuit Breaker	**104X100FA001**	1	
Circuit Breaker	**104X100FA002**		1
SCR's	**104X125DA042**	4	
SCR's	**104X125DA113**		4
Fuse 5FU and 6FU 5 amps	**104X109AD035** (Bussman Type ABC-5) (Littlefuse Type 314005)	2	2

WARRANTY

The Contract Warranty for Statotrol drives is stated in the General Electric Apparatus Handbook, Section 98, Page 11. The purpose of the following is to provide specific instructions to the Statotrol user regarding warranty administration and how to obtain assistance on out-of-warranty failures.

STATOTROL CONTROL (3 & 5 HP)

The warranty covers all major parts of the controller such as the main printed circuit board, modification kits, contactors, etc., but does not provide for replacement of fuses or the complete controller.

1. In the event of failure or misapplication during "in warranty" refer to the instruction book to identify the defective part or subassembly.

2. When the defective part has been identified (or for assistance in identification) call:

STATOTROL DRIVE SERVICE
Erie, Pennsylvania
(814) 455-0521

BEFORE CALLING, LIST CATALOG NUMBERS OF THE CONTROLLER, MOTOR OPERATOR'S STATION AND ANY OPTIONAL MODIFICATIONS.
INCLUDE ORANGE COLORED IDENTIFICATION NUMBER LOCATED IN LOWER CORNER OF CONTROLLER CASE.

3. Statotrol Service will ship prepaid, at no charge any parts that have failed in warranty and advise if failed part(s) should be returned to factory. The factory address is:

GENERAL ELECTRIC
SPEED VARIATOR PRODUCTS OPERATION
1100 LAWRENCE PARKWAY
ERIE, PENNSYLVANIA 16531
ATTENTION: STATOTROL DRIVES SERVICE

4. For parts, contact your local General Electric Sales Office or your "Authorized" General Electric Parts Distributor.

"Exchange" parts at "Exchange" prices may be available for selected items. When returning parts for exchange, pack them carefully to prevent damage in transit.

STATOTROL MOTORS

Statotrol motor repairs are generally handled by General Electric Small Motor Service Stations or Apparatus Service Shops. For specific instructions on your motor, call STATOTROL DRIVE SERVICE at the number listed above and furnish complete nameplate data.

GENERAL ELECTRIC COMPANY
SPEED VARIATOR PRODUCTS OPERATION
ERIE, PENNSYLVANIA 16531

GENERAL ELECTRIC